

23. Multi-Function Pin Controller (MPC)

23.1 Overview

The multi-function pin controller (MPC) selects and assigns input/output of peripheral functions and interrupt input signals from multiple ports. The MPC also assigns the port of external bus related signals.

Table 23.1 lists the functions assigned to each multiplexed pin. The symbols ✓ and × in the table indicate whether the pin is available or unavailable for the package. Selecting a single function for multiple pins is prohibited.

Table 23.1 Functions Assigned to Each Multiplexed Pin (1/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Interrupt		NMI (input)	P35	✓	✓	✓	✓
EXDMA controller	EXDMAC0	EDREQ0 (input)	P22	✓	✓	✓	✓
			P55	✓	✓	✓	✓
			P80	✓	✓	✓	x
		EDACK0 (output)	P54	✓	✓	✓	✓
			P23	✓	✓	✓	✓
			P81	✓	✓	✓	x
	EXDMAC1	EDREQ1 (input)	P24	✓	✓	✓	✓
			P33	✓	✓	✓	✓
			P82	✓	✓	✓	x
		EDACK1 (output)	P25	✓	✓	✓	✓
			P56	✓	✓	✓	x
			P83	✓	✓	✓	x
Interrupt	IRQ0	IRQ0-DS (input)	P30	✓	✓	✓	✓
		IRQ0 (input)	P10	✓	✓	x	x
			PD0	✓	✓	✓	✓
	IRQ1	IRQ1-DS (input)	P31	✓	✓	✓	✓
		IRQ1 (input)	P11	✓	✓	x	x
			PD1	✓	✓	✓	✓
	IRQ2	IRQ2-DS (input)	P32	✓	✓	✓	✓
		IRQ2 (input)	P12	✓	✓	✓	✓
			PD2	✓	✓	✓	✓
	IRQ3	IRQ3-DS (input)	P33	✓	✓	✓	✓
		IRQ3 (input)	P13	✓	✓	✓	✓
			PD3	✓	✓	✓	✓
	IRQ4	IRQ4-DS (input)	PB1	✓	✓	✓	✓
		IRQ4 (input)	P14	✓	✓	✓	✓
			P34	✓	✓	✓	✓
			PD4	✓	✓	✓	✓
			PF5	✓	✓	✓	x
	IRQ5	IRQ5-DS (input)	PA4	✓	✓	✓	✓
		IRQ5 (input)	P15	✓	✓	✓	✓
			PD5	✓	✓	✓	✓
			PE5	✓	✓	✓	✓
	IRQ6	IRQ6-DS (input)	PA3	✓	✓	✓	✓
		IRQ6 (input)	P16	✓	✓	✓	✓
			PD6	✓	✓	✓	✓
			PE6	✓	✓	✓	✓
	IRQ7	IRQ7-DS (input)	PE2	✓	✓	✓	✓
		IRQ7 (input)	P17	✓	✓	✓	✓
			PD7	✓	✓	✓	✓
			PE7	✓	✓	✓	✓

Table 23.1 Functions Assigned to Each Multiplexed Pin (2/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Interrupt	IRQ8	IRQ8-DS (input)	P40	✓	✓	✓	✓
		IRQ8 (input)	P00	✓	✓	✓	×
			P20	✓	✓	✓	✓
	IRQ9	IRQ9-DS (input)	P41	✓	✓	✓	✓
		IRQ9 (input)	P01	✓	✓	✓	×
			P21	✓	✓	✓	✓
	IRQ10	IRQ10-DS (input)	P42	✓	✓	✓	✓
		IRQ10 (input)	P02	✓	✓	✓	×
			P55	✓	✓	✓	✓
	IRQ11	IRQ11-DS (input)	P43	✓	✓	✓	✓
		IRQ11 (input)	P03	✓	✓	✓	×
			PA1	✓	✓	✓	✓
	IRQ12	IRQ12-DS (input)	P44	✓	✓	✓	✓
		IRQ12 (input)	PB0	✓	✓	✓	✓
			PC1	✓	✓	✓	✓
	IRQ13	IRQ13-DS (input)	P45	✓	✓	✓	✓
		IRQ13 (input)	P05	✓	✓	✓	✓
			PC6	✓	✓	✓	✓
	IRQ14	IRQ14-DS (input)	P46	✓	✓	✓	✓
		IRQ14 (input)	PC0	✓	✓	✓	✓
			PC7	✓	✓	✓	✓
	IRQ15	IRQ15-DS (input)	P47	✓	✓	✓	✓
		IRQ15 (input)	P07	✓	✓	✓	✓
			P67	✓	✓	✓	×
Multi-function timer unit 3	MTU0	MTIOC0A (input/output)	P34	✓	✓	✓	✓
			PB3	✓	✓	✓	✓
		MTIOC0B (input/output)	P13	✓	✓	✓	✓
			P15	✓	✓	✓	✓
			PA1	✓	✓	✓	✓
		MTIOC0C (input/output)	P32	✓	✓	✓	✓
			PB1	✓	✓	✓	✓
		MTIOC0D (input/output)	P33	✓	✓	✓	✓
			PA3	✓	✓	✓	✓
	MTU1	MTIOC1A (input/output)	P20	✓	✓	✓	✓
			PE4	✓	✓	✓	✓
		MTIOC1B (input/output)	P21	✓	✓	✓	✓
			PB5	✓	✓	✓	✓
	MTU2	MTIOC2A (input/output)	P26	✓	✓	✓	✓
			PB5	✓	✓	✓	✓
		MTIOC2B (input/output)	P27	✓	✓	✓	✓
			PE5	✓	✓	✓	✓

Table 23.1 Functions Assigned to Each Multiplexed Pin (3/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Multi-function timer unit 3	MTU3	MTIOC3A (input/output)	P14	✓	✓	✓	✓
			P17	✓	✓	✓	✓
			PC1	✓	✓	✓	✓
			PC7	✓	✓	✓	✓
		MTIOC3B (input/output)	P17	✓	✓	✓	✓
			P22	✓	✓	✓	✓
			P80	✓	✓	✓	×
			PB7	✓	✓	✓	✓
			PC5	✓	✓	✓	✓
			PE1	✓	✓	✓	✓
		MTIOC3C (input/output)	P16	✓	✓	✓	✓
			P56	✓	✓	✓	×
			PC0	✓	✓	✓	✓
			PC6	✓	✓	✓	✓
			PJ3	✓	✓	✓	✓
		MTIOC3D (input/output)	P16	✓	✓	✓	✓
			P23	✓	✓	✓	✓
			P81	✓	✓	✓	×
			PB6	✓	✓	✓	✓
			PC4	✓	✓	✓	✓
			PE0	✓	✓	✓	✓
	MTU4	MTIOC4A (input/output)	P21	✓	✓	✓	✓
			P24	✓	✓	✓	✓
			P82	✓	✓	✓	×
			PA0	✓	✓	✓	✓
			PB3	✓	✓	✓	✓
			PE2	✓	✓	✓	✓
		MTIOC4B (input/output)	P17	✓	✓	✓	✓
			P30	✓	✓	✓	✓
			P54	✓	✓	✓	✓
			PC2	✓	✓	✓	✓
			PD1	✓	✓	✓	✓
			PE3	✓	✓	✓	✓
		MTIOC4C (input/output)	P25	✓	✓	✓	✓
			P83	✓	✓	✓	×
			P87	✓	✓	✓	×
			PB1	✓	✓	✓	✓
			PE1	✓	✓	✓	✓
			PE5	✓	✓	✓	✓

Table 23.1 Functions Assigned to Each Multiplexed Pin (4/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Multi-function timer unit 3	MTU4	MTIOC4D (input/output)	P31	✓	✓	✓	✓
			P55	✓	✓	✓	✓
			P86	✓	✓	✓	×
			PC3	✓	✓	✓	✓
			PD2	✓	✓	✓	✓
			PE4	✓	✓	✓	✓
	MTU5	MTIC5U (input)	P12	✓	✓	×	×
			PA4	✓	✓	✓	✓
			PD7	✓	✓	✓	✓
		MTIC5V (input)	P11	✓	✓	×	×
			PA6	✓	✓	✓	✓
			PD6	✓	✓	✓	✓
		MTIC5W (input)	P10	✓	✓	×	×
			PB0	✓	✓	✓	✓
			PD5	✓	✓	✓	✓
	MTU6	MTIOC6A (input/output)	PE7	✓	✓	✓	✓
			PJ1	✓	✓	×	×
		MTIOC6B (input/output)	PA5	✓	✓	✓	✓
			PJ0	✓	✓	×	×
		MTIOC6C (input/output)	P85	✓	✓	×	×
			PE6	✓	✓	✓	✓
		MTIOC6D (input/output)	P84	✓	✓	×	×
			PA0	✓	✓	✓	✓
	MTU7	MTIOC7A (input/output)	PA2	✓	✓	✓	✓
		MTIOC7B (input/output)	PA1	✓	✓	✓	✓
		MTIOC7C (input/output)	P67	✓	✓	✓	×
		MTIOC7D (input/output)	P66	✓	✓	✓	×
	MTU8	MTIOC8A (input/output)	PD6	✓	✓	✓	✓
		MTIOC8B (input/output)	PD4	✓	✓	✓	✓
		MTIOC8C (input/output)	PD5	✓	✓	✓	✓
		MTIOC8D (input/output)	PD3	✓	✓	✓	✓
	MTU	MTCLKA (input)	P14	✓	✓	✓	✓
			P24	✓	✓	✓	✓
			PA4	✓	✓	✓	✓
			PC6	✓	✓	✓	✓
			PD5	✓	✓	✓	✓
		MTCLKB (input)	P15	✓	✓	✓	✓
			P25	✓	✓	✓	✓
			PA6	✓	✓	✓	✓
			PC7	✓	✓	✓	✓
		MTCLKC (input)	P22	✓	✓	✓	✓
			PA1	✓	✓	✓	✓
			PC4	✓	✓	✓	✓

Table 23.1 Functions Assigned to Each Multiplexed Pin (5/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Multi-function timer unit 3	MTU	MTCLKD (input)	P23	✓	✓	✓	✓
			PA3	✓	✓	✓	✓
			PC5	✓	✓	✓	✓
Port output enable 3	POE0	POE0# (input)	P32	✓	✓	✓	✓
			P93	✓	✓	✓	×
			PC4	✓	✓	✓	✓
			PD1	✓	✓	✓	✓
			PD7	✓	✓	✓	✓
	POE4	POE4# (input)	P33	✓	✓	✓	✓
			P92	✓	✓	✓	×
			PB5	✓	✓	✓	✓
			PD0	✓	✓	✓	✓
			PD6	✓	✓	✓	✓
	POE8	POE8# (input)	P17	✓	✓	✓	✓
			P30	✓	✓	✓	✓
			PD3	✓	✓	✓	✓
			PE3	✓	✓	✓	✓
			PJ5	✓	✓	✓	×
	POE10	POE10# (input)	P32	✓	✓	✓	✓
			P34	✓	✓	✓	✓
			PA6	✓	✓	✓	✓
			PD5	✓	✓	✓	✓
	POE11	POE11# (input)	P33	✓	✓	✓	✓
			PB3	✓	✓	✓	✓
			PD4	✓	✓	✓	✓
General PWM timer W	GPTW	GTADSM0 (output)	P12	✓	✓	✓	✓
			PH4	✓	×	×	×
			PK4	✓	×	×	×
			PL4	✓	×	×	×
			PM4	✓	×	×	×
		GTADSM1 (output)	P13	✓	✓	✓	✓
			PH5	✓	×	×	×
			PK5	✓	×	×	×
			PL5	✓	×	×	×
			PM5	✓	×	×	×
		GTETRGA (input)	P15	✓	✓	✓	✓
			PH0	✓	×	×	×
			PK0	✓	×	×	×
			PL0	✓	×	×	×
			PM0	✓	×	×	×

Table 23.1 Functions Assigned to Each Multiplexed Pin (6/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
General PWM timer W	GPTW	GTETRGB (input)	PA6	✓	✓	✓	✓
			PH1	✓	x	x	x
			PK1	✓	x	x	x
			PL1	✓	x	x	x
			PM1	✓	x	x	x
		GTETRG (input)	PC4	✓	✓	✓	✓
			PH2	✓	x	x	x
			PK2	✓	x	x	x
			PL2	✓	x	x	x
			PM2	✓	x	x	x
		GTETRGD (input)	P14	✓	✓	✓	✓
			PH3	✓	x	x	x
			PK3	✓	x	x	x
			PL3	✓	x	x	x
			PM3	✓	x	x	x
	GPTW0	GTIOC0A (input/output)	P23	✓	✓	✓	✓
			P83	✓	✓	✓	x
			PA5	✓	✓	✓	✓
			PD3	✓	✓	✓	✓
			PE5	✓	✓	✓	✓
			PH6	✓	x	x	x
		GTIOC0B (input/output)	P17	✓	✓	✓	✓
			P81	✓	✓	✓	x
			PA0	✓	✓	✓	✓
			PD2	✓	✓	✓	✓
			PE2	✓	✓	✓	✓
			PH7	✓	x	x	x
	GPTW1	GTIOC1A (input/output)	P22	✓	✓	✓	✓
			PA2	✓	✓	✓	✓
			PC5	✓	✓	✓	✓
			PD1	✓	✓	✓	✓
			PE4	✓	✓	✓	✓
			PK6	✓	x	x	x
		GTIOC1B (input/output)	P67	✓	✓	✓	x
			P87	✓	✓	✓	x
			PC3	✓	✓	✓	✓
			PD0	✓	✓	✓	✓

Table 23.1 Functions Assigned to Each Multiplexed Pin (7/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
General PWM timer W	GPTW2	GTIOC2A (input/output)	P21	✓	✓	✓	✓
			P82	✓	✓	✓	×
			PA1	✓	✓	✓	✓
			PE3	✓	✓	✓	✓
			PL6	✓	×	×	×
		GTIOC2B (input/output)	P66	✓	✓	✓	×
			P86	✓	✓	✓	×
			PC2	✓	✓	✓	✓
			PE0	✓	✓	✓	✓
			PL7	✓	×	×	×
	GPTW3	GTIOC3A (input/output)	PC7	✓	✓	✓	✓
			PE7	✓	✓	✓	✓
			PM6	✓	×	×	×
		GTIOC3B (input/output)	PC6	✓	✓	✓	✓
			PE6	✓	✓	✓	✓
			PM7	✓	×	×	×
16-bit timer pulse unit	TPU0	TIOCA0 (input/output)	P86	✓	✓	✓	×
			PA0	✓	✓	✓	✓
		TIOCB0 (input/output)	P17	✓	✓	✓	✓
			PA1	✓	✓	✓	✓
		TIOCC0 (input/output)	P32	✓	✓	✓	✓
			P85	✓	✓	×	×
		TIOCD0 (input/output)	P33	✓	✓	✓	✓
			PA3	✓	✓	✓	✓
	TPU1	TIOCA1 (input/output)	P56	✓	✓	✓	×
			PA4	✓	✓	✓	✓
		TIOCB1 (input/output)	P16	✓	✓	✓	✓
			PA5	✓	✓	✓	✓
	TPU2	TIOCA2 (input/output)	P87	✓	✓	✓	×
			PA6	✓	✓	✓	✓
		TIOCB2 (input/output)	P15	✓	✓	✓	✓
			PA7	✓	✓	✓	✓
	TPU3	TIOCA3 (input/output)	P21	✓	✓	✓	✓
			PB0	✓	✓	✓	✓
		TIOCB3 (input/output)	P20	✓	✓	✓	✓
			PB1	✓	✓	✓	✓
		TIOCC3 (input/output)	P22	✓	✓	✓	✓
			PB2	✓	✓	✓	✓
		TIOCD3 (input/output)	P23	✓	✓	✓	✓
			PB3	✓	✓	✓	✓
	TPU4	TIOCA4 (input/output)	P25	✓	✓	✓	✓
			PB4	✓	✓	✓	✓
		TIOCB4 (input/output)	P24	✓	✓	✓	✓
			PB5	✓	✓	✓	✓

Table 23.1 Functions Assigned to Each Multiplexed Pin (8/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
16-bit timer pulse unit	TPU5	TIOCA5 (input/output)	P13	✓	✓	✓	✓
			PB6	✓	✓	✓	✓
		TIOCB5 (input/output)	P14	✓	✓	✓	✓
			PB7	✓	✓	✓	✓
	TPU (unit 0)	TCLKA (input)	P14	✓	✓	✓	✓
			PC2	✓	✓	✓	✓
		TCLKB (input)	P15	✓	✓	✓	✓
			PA3	✓	✓	✓	✓
			PC3	✓	✓	✓	✓
		TCLKC (input)	P16	✓	✓	✓	✓
			PB2	✓	✓	✓	✓
			PC0	✓	✓	✓	✓
		TCLKD (input)	P17	✓	✓	✓	✓
			PB3	✓	✓	✓	✓
			PC1	✓	✓	✓	✓
Programmable pulse generator	PPG0	PO0 (output)	P20	✓	✓	✓	✓
		PO1 (output)	P21	✓	✓	✓	✓
		PO2 (output)	P22	✓	✓	✓	✓
		PO3 (output)	P23	✓	✓	✓	✓
		PO4 (output)	P24	✓	✓	✓	✓
		PO5 (output)	P25	✓	✓	✓	✓
		PO6 (output)	P26	✓	✓	✓	✓
		PO7 (output)	P27	✓	✓	✓	✓
		PO8 (output)	P30	✓	✓	✓	✓
		PO9 (output)	P31	✓	✓	✓	✓
		PO10 (output)	P32	✓	✓	✓	✓
		PO11 (output)	P33	✓	✓	✓	✓
		PO12 (output)	P34	✓	✓	✓	✓
		PO13 (output)	P13	✓	✓	✓	✓
			P15	✓	✓	✓	✓
		PO14 (output)	P16	✓	✓	✓	✓
		PO15 (output)	P14	✓	✓	✓	✓
			P17	✓	✓	✓	✓
	PPG1	PO16 (output)	P73	✓	✓	✓	×
			PA0	✓	✓	✓	✓
		PO17 (output)	PA1	✓	✓	✓	✓
			PC0	✓	✓	✓	✓
			PE1	✓	✓	✓	✓
		PO18 (output)	PA2	✓	✓	✓	✓
			PC1	✓	✓	✓	✓
			PE1	✓	✓	✓	✓
		PO19 (output)	P74	✓	✓	✓	×
			PA3	✓	✓	✓	✓
		PO20 (output)	P75	✓	✓	✓	×
			PA4	✓	✓	✓	✓

Table 23.1 Functions Assigned to Each Multiplexed Pin (9/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Programmable pulse generator	PPG1	PO21 (output)	PA5	✓	✓	✓	✓
			PC2	✓	✓	✓	✓
		PO22 (output)	P76	✓	✓	✓	×
			PA6	✓	✓	✓	✓
		PO23 (output)	P77	✓	✓	✓	×
			PA7	✓	✓	✓	✓
			PE2	✓	✓	✓	✓
		PO24 (output)	PB0	✓	✓	✓	✓
			PC3	✓	✓	✓	✓
		PO25 (output)	PB1	✓	✓	✓	✓
			PC4	✓	✓	✓	✓
		PO26 (output)	P80	✓	✓	✓	×
			PB2	✓	✓	✓	✓
			PE3	✓	✓	✓	✓
		PO27 (output)	P81	✓	✓	✓	×
			PB3	✓	✓	✓	✓
		PO28 (output)	P82	✓	✓	✓	×
			PB4	✓	✓	✓	✓
			PE4	✓	✓	✓	✓
		PO29 (output)	PB5	✓	✓	✓	✓
			PC5	✓	✓	✓	✓
		PO30 (output)	PB6	✓	✓	✓	✓
			PC6	✓	✓	✓	✓
		PO31 (output)	PB7	✓	✓	✓	✓
			PC7	✓	✓	✓	✓
8-bit timer	TMR0	TMO0 (output)	P22	✓	✓	✓	✓
			PB3	✓	✓	✓	✓
		TMC10 (input)	P01	✓	✓	✓	×
			P21	✓	✓	✓	✓
			PB1	✓	✓	✓	✓
		TMR10 (input)	P00	✓	✓	✓	×
			P20	✓	✓	✓	✓
			PA4	✓	✓	✓	✓
	TMR1	TMO1 (output)	P17	✓	✓	✓	✓
			P26	✓	✓	✓	✓
		TMC11 (input)	P02	✓	✓	✓	×
			P12	✓	✓	✓	✓
			P54	✓	✓	✓	✓
			PC4	✓	✓	✓	✓
		TMR11 (input)	P24	✓	✓	✓	✓
			PB5	✓	✓	✓	✓
	TMR2	TMO2 (output)	P16	✓	✓	✓	✓
			PC7	✓	✓	✓	✓

Table 23.1 Functions Assigned to Each Multiplexed Pin (10/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
8-bit timer	TMR2	TMC12 (input)	P15	✓	✓	✓	✓
			P31	✓	✓	✓	✓
			PC6	✓	✓	✓	✓
		TMRI2 (input)	P14	✓	✓	✓	✓
			PC5	✓	✓	✓	✓
	TMR3	TMO3 (output)	P13	✓	✓	✓	✓
			P32	✓	✓	✓	✓
			P55	✓	✓	✓	✓
		TMC13 (input)	P11	✓	✓	x	x
			P27	✓	✓	✓	✓
			P34	✓	✓	✓	✓
			PA6	✓	✓	✓	✓
		TMRI3 (input)	P10	✓	✓	x	x
			P30	✓	✓	✓	✓
			P33	✓	✓	✓	✓
Compare match timer W	CMTW0	TOC0 (output)	PC7	✓	✓	✓	✓
			PH1	✓	x	x	x
		TIC0 (input)	PC6	✓	✓	✓	✓
			PH0	✓	x	x	x
		TOC1 (output)	PE7	✓	✓	✓	✓
			PK1	✓	x	x	x
		TIC1 (input)	PE6	✓	✓	✓	✓
			PK0	✓	x	x	x
	CMTW1	TOC2 (output)	PD3	✓	✓	✓	✓
			PL1	✓	x	x	x
		TIC2 (input)	PD2	✓	✓	✓	✓
			PL0	✓	x	x	x
		TOC3 (output)	PE3	✓	✓	✓	✓
			PM1	✓	x	x	x
		TIC3 (input)	PE2	✓	✓	✓	✓
			PM0	✓	x	x	x
Realtime clock		RTCOUT (output)	P16	✓	✓	✓	✓
			P32	✓	✓	✓	✓
		RTCIC0 (input)*1	P30	✓	✓	✓	✓
		RTCIC1 (input)*1	P31	✓	✓	✓	✓
		RTCIC2 (input)*1	P32	✓	✓	✓	✓
PTP module for the ethernet controller		EPLSOUT0 (output)	P17	✓	✓	✓	✓
			PJ0	✓	✓	x	x
			PJ5	✓	✓	✓	x
		EPLSOUT1 (output)	P11	✓	✓	x	x
			P67	✓	✓	✓	x
			P87	✓	✓	✓	x
			PJ1	✓	✓	x	x

Table 23.1 Functions Assigned to Each Multiplexed Pin (11/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Ethernet controller	RMII0	REF50CK0 (input)	P76	✓	✓	✓	×
			PB2	✓	✓	✓	✓
			PE5	✓	✓	✓	✓
			PL3	✓	×	×	×
		RMII0_CRS_DV (input)	P83	✓	✓	✓	×
			PB7	✓	✓	✓	✓
			PM7	✓	×	×	×
		RMII0_TXD0 (output)	P81	✓	✓	✓	×
			PB5	✓	✓	✓	✓
			PL4	✓	×	×	×
		RMII0_TXD1 (output)	P82	✓	✓	✓	×
			PB6	✓	✓	✓	✓
			PL5	✓	×	×	×
		RMII0_RXD0 (input)	P75	✓	✓	✓	×
			PB1	✓	✓	✓	✓
			PL0	✓	×	×	×
		RMII0_RXD1 (input)	P74	✓	✓	✓	×
			PB0	✓	✓	✓	✓
			PL1	✓	×	×	×
		RMII0_TXD_EN (output)	P80	✓	✓	✓	×
			PA0	✓	✓	✓	✓
			PB4	✓	✓	✓	✓
			PL6	✓	×	×	×
		RMII0_RX_ER (input)	P77	✓	✓	✓	×
			PB3	✓	✓	✓	✓
			PL2	✓	×	×	×

Table 23.1 Functions Assigned to Each Multiplexed Pin (12/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Ethernet controller	RMII1	REF50CK1 (input)	PD6	✓	✓	✓	x
			PG0	✓	✓	x	x
			PQ4	✓	x	x	x
		RMII1_CRS_DV (input)	P92	✓	✓	✓	x
			PQ0	✓	x	x	x
		RMII1_TXD0 (output)	P64	✓	✓	✓	x
			PG3	✓	✓	x	x
			PQ5	✓	x	x	x
		RMII1_TXD1 (output)	P63	✓	✓	✓	x
			PG4	✓	✓	x	x
			PQ6	✓	x	x	x
		RMII1_RXD0 (input)	P62	✓	✓	✓	x
			P94	✓	✓	x	x
			PM0	✓	x	x	x
		RMII1_RXD1 (input)	P61	✓	✓	✓	x
			P95	✓	✓	x	x
			PM1	✓	x	x	x
		RMII1_TXD_EN (output)	P60	✓	✓	✓	x
			PQ7	✓	x	x	x
		RMII1_RX_ER (input)	PD7	✓	✓	✓	x
			PG1	✓	✓	x	x
			PN3	✓	x	x	x

Table 23.1 Functions Assigned to Each Multiplexed Pin (13/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Ethernet controller	MII0	ET0_CRS (input)	P83	✓	✓	✓	×
			PB7	✓	✓	✓	✓
			PM7	✓	×	×	×
		ET0_RX_DV (input)	PC2	✓	✓	✓	✓
			PK2	✓	×	×	×
		ET0_EXOUT (output)	P55	✓	✓	✓	✓
			PA6	✓	✓	✓	✓
			PJ3	✓	✓	✓	✓
		ET0_LINKSTA (input)	P34	✓	✓	✓	✓
			P54	✓	✓	✓	✓
			PA5	✓	✓	✓	✓
		ET0_ETXD0 (output)	P81	✓	✓	✓	×
			PB5	✓	✓	✓	✓
			PL4	✓	×	×	×
		ET0_ETXD1 (output)	P82	✓	✓	✓	×
			PB6	✓	✓	✓	✓
			PL5	✓	×	×	×
		ET0_ETXD2 (output)	PC5	✓	✓	✓	✓
			PM4	✓	×	×	×
		ET0_ETXD3 (output)	PC6	✓	✓	✓	✓
			PM5	✓	×	×	×
		ET0_ERXD0 (input)	P75	✓	✓	✓	×
			PB1	✓	✓	✓	✓
			PL0	✓	×	×	×
		ET0_ERXD1 (input)	P74	✓	✓	✓	×
			PB0	✓	✓	✓	✓
			PL1	✓	×	×	×
		ET0_ERXD2 (input)	PC1	✓	✓	✓	✓
			PE4	✓	✓	✓	✓
			PK4	✓	×	×	×
		ET0_ERXD3 (input)	PC0	✓	✓	✓	✓
			PE3	✓	✓	✓	✓
			PK5	✓	×	×	×
		ET0_TX_EN (output)	P80	✓	✓	✓	×
			PA0	✓	✓	✓	✓
			PB4	✓	✓	✓	✓
			PL6	✓	×	×	×
		ET0_TX_ER (output)	PC3	✓	✓	✓	✓
			PK3	✓	×	×	×
		ET0_RX_ER (input)	P77	✓	✓	✓	×
			PB3	✓	✓	✓	✓
			PL2	✓	×	×	×
		ET0_TX_CLK (input)	PC4	✓	✓	✓	✓
			PM6	✓	×	×	×

Table 23.1 Functions Assigned to Each Multiplexed Pin (14/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Ethernet controller	MII0	ET0_RX_CLK (input)	P76	✓	✓	✓	×
			PB2	✓	✓	✓	✓
			PE5	✓	✓	✓	✓
			PL3	✓	×	×	×
		ET0_COL (input)	PC7	✓	✓	✓	✓
			PK1	✓	×	×	×
		ET0_WOL (output)	P73	✓	✓	✓	×
			PA1	✓	✓	✓	✓
			PA7	✓	✓	✓	✓
		ET0_MDC (output)	P72	✓	✓	✓	×
			PA4	✓	✓	✓	✓
			PK0	✓	×	×	×
		ET0_MDIO (input/output)	P71	✓	✓	✓	×
			PA3	✓	✓	✓	✓
			PL7	✓	×	×	×

Table 23.1 Functions Assigned to Each Multiplexed Pin (15/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Ethernet controller	MII1	ET1_CRS (input)	P92	✓	✓	x	x
			PQ0	✓	x	x	x
		ET1_RX_DV (input)	P90	✓	✓	x	x
			PQ2	✓	x	x	x
		ET1_EXOUT (output)	P26	✓	✓	✓	x
			PD2	✓	✓	✓	x
		ET1_LINKSTA (input)	P84	✓	✓	x	x
			P93	✓	✓	✓	x
		ET1_ETXD0 (output)	P64	✓	✓	x	x
			PG3	✓	✓	x	x
			PQ5	✓	x	x	x
		ET1_ETXD1 (output)	P63	✓	✓	x	x
			PG4	✓	✓	x	x
			PQ6	✓	x	x	x
		ET1_ETXD2 (output)	PG5	✓	✓	x	x
			PN0	✓	x	x	x
		ET1_ETXD3 (output)	PG6	✓	✓	x	x
			PN1	✓	x	x	x
		ET1_ERXD0 (input)	P62	✓	✓	x	x
			P94	✓	✓	x	x
			PM0	✓	x	x	x
		ET1_ERXD1 (input)	P61	✓	✓	x	x
			P95	✓	✓	x	x
			PM1	✓	x	x	x
		ET1_ERXD2 (input)	P96	✓	✓	x	x
			PM2	✓	x	x	x
		ET1_ERXD3 (input)	P97	✓	✓	x	x
			PM3	✓	x	x	x
		ET1_TX_EN (output)	P60	✓	✓	x	x
			PQ7	✓	x	x	x
		ET1_TX_ER (output)	PG7	✓	✓	x	x
			PQ3	✓	x	x	x
		ET1_RX_ER (input)	PD7	✓	✓	x	x
			PG1	✓	✓	x	x
			PN3	✓	x	x	x
		ET1_TX_CLK (input)	PG2	✓	✓	x	x
			PN2	✓	x	x	x
		ET1_RX_CLK (input)	PD6	✓	✓	x	x
			PG0	✓	✓	x	x
			PQ4	✓	x	x	x

Table 23.1 Functions Assigned to Each Multiplexed Pin (16/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Ethernet controller	MII1	ET1_COL (input)	P91	✓	✓	×	×
			PQ1	✓	×	×	×
		ET1_WOL (output)	P27	✓	✓	✓	×
			PD3	✓	✓	✓	×
		ET1_MDC (output)	P31	✓	✓	✓	×
			PD5	✓	✓	✓	×
			PN5	✓	×	×	×
		ET1_MDIO (input/output)	P30	✓	✓	✓	×
			PD4	✓	✓	✓	×
			PN4	✓	×	×	×
PHY management interface	PMGI0	PMGI0_MDC (output)	P72	✓	✓	✓	×
			PA4	✓	✓	✓	✓
			PK0	✓	×	×	×
		PMGI0_MDIO (input/output)	P71	✓	✓	✓	×
			PA3	✓	✓	✓	✓
			PL7	✓	×	×	×
	PMGI1	PMGI1_MDC (output)	P31	✓	✓	✓	×
			PD5	✓	✓	✓	×
			PN5	✓	×	×	×
		PMGI1_MDIO (input/output)	P30	✓	✓	✓	×
			PD4	✓	✓	✓	×
			PN4	✓	×	×	×

Table 23.1 Functions Assigned to Each Multiplexed Pin (17/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Serial communications interface	SCI0	CTS0# (input)/RTS0# (output)/SS0# (input)	P23	✓	✓	✓	✓
			PJ3	✓	✓	✓	✓
		RXD0 (input)/SMISO0 (input/output)/SSCL0 (input/output)	P21	✓	✓	✓	✓
			P33	✓	✓	✓	✓
		SCK0 (input/output)	P22	✓	✓	✓	✓
			P34	✓	✓	✓	✓
		SMOSI0 (input/output)/SSDA0 (input/output)/TXD0 (output)	P20	✓	✓	✓	✓
			P32	✓	✓	✓	✓
	SCI1	CTS1# (input)/RTS1# (output)/SS1# (input)	P14	✓	✓	✓	✓
			P31	✓	✓	✓	✓
		RXD1 (input)/SMISO1 (input/output)/SSCL1 (input/output)	P15	✓	✓	✓	✓
			P30	✓	✓	✓	✓
			PF2	✓	✓	x	x
		SCK1 (input/output)	P17	✓	✓	✓	✓
			P27	✓	✓	✓	✓
			PF1	✓	✓	x	x
		SMOSI1 (input/output)/SSDA1 (input/output)/TXD1 (output)	P16	✓	✓	✓	✓
			P26	✓	✓	✓	✓
			PF0	✓	✓	x	x
	SCI2	CTS2# (input)/RTS2# (output)/SS2# (input)	P54	✓	✓	✓	✓
			PJ5	✓	✓	✓	x
		RXD2 (input)/SMISO2 (input/output)/SSCL2 (input/output)	P12	✓	✓	✓	✓
			P52	✓	✓	✓	✓
		SCK2 (input/output)	P11	✓	✓	x	x
			P51	✓	✓	✓	✓
		SMOSI2 (input/output)/SSDA2 (input/output)/TXD2 (output)	P13	✓	✓	✓	✓
			P50	✓	✓	✓	✓
	SCI3	CTS3# (input)/RTS3# (output)/SS3# (input)	P26	✓	✓	✓	✓
		RXD3 (input)/SMISO3 (input/output)/SSCL3 (input/output)	P16	✓	✓	✓	✓
			P25	✓	✓	✓	✓
		SCK3 (input/output)	P15	✓	✓	✓	✓
			P24	✓	✓	✓	✓
		SMOSI3 (input/output)/SSDA3 (input/output)/TXD3 (output)	P17	✓	✓	✓	✓
			P23	✓	✓	✓	✓
	SCI4	CTS4# (input)/RTS4# (output)/SS4# (input)	PB2	✓	✓	✓	x
		RXD4 (input)/SMISO4 (input/output)/SSCL4 (input/output)	PB0	✓	✓	✓	x
		SCK4 (input/output)	PB3	✓	✓	✓	x
		SMOSI4 (input/output)/SSDA4 (input/output)/TXD4 (output)	PB1	✓	✓	✓	x

Table 23.1 Functions Assigned to Each Multiplexed Pin (18/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Serial communications interface	SCI5	CTS5# (input)/RTS5# (output)/SS5# (input)	PA6	✓	✓	✓	✓
			PC0	✓	✓	✓	✓
		RXD5 (input)/SMISO5 (input/output)/SSCL5 (input/output)	PA2	✓	✓	✓	✓
			PA3	✓	✓	✓	✓
			PC2	✓	✓	✓	✓
			PC4	✓	✓	✓	✓
		SCK5 (input/output)	PA1	✓	✓	✓	✓
			PC1	✓	✓	✓	✓
			PC4	✓	✓	✓	✓
		SMOSI5 (input/output)/SSDA5 (input/output)/TXD5 (output)	PA4	✓	✓	✓	✓
			PC3	✓	✓	✓	✓
	SCI6	CTS6# (input)/RTS6# (output)/SS6# (input)	PB2	✓	✓	✓	✓
			PJ3	✓	✓	✓	✓
		RXD6 (input)/SMISO6 (input/output)/SSCL6 (input/output)	P01	✓	✓	✓	×
			P33	✓	✓	✓	✓
			PB0	✓	✓	✓	✓
			PB3	✓	✓	✓	✓
		SCK6 (input/output)	P02	✓	✓	✓	×
			P34	✓	✓	✓	✓
			PB3	✓	✓	✓	✓
		SMOSI6 (input/output)/SSDA6 (input/output)/TXD6 (output)	P00	✓	✓	✓	×
			P32	✓	✓	✓	✓
			PB1	✓	✓	✓	✓
	SCI7	CTS7# (input)/RTS7# (output)/SS7# (input)	P93	✓	✓	✓	×
			PH3	✓	×	×	×
		RXD7 (input)/SMISO7 (input/output)/SSCL7 (input/output)	P57	✓	✓	×	×
			P92	✓	✓	✓	×
			PH1	✓	×	×	×
			PH0	✓	×	×	×
		SCK7 (input/output)	P56	✓	✓	✓	×
			P91	✓	✓	✓	×
			PH0	✓	×	×	×
		SMOSI7 (input/output)/SSDA7 (input/output)/TXD7 (output)	P55	✓	✓	✓	×
			P90	✓	✓	✓	×
			PH2	✓	×	×	×
	SCI8	CTS8# (input)/SS8# (input)	PC4	✓	✓	✓	✓
			PK3	✓	×	×	×
		RTS8# (output)/CTS8# (input)/SS8# (input)	PC5	✓	✓	✓	✓
			PC6	✓	✓	✓	✓
		RXD8 (input)/SMISO8 (input/output)/SSCL8 (input/output)	PJ1	✓	✓	×	×
			PK1	✓	×	×	×
			PK0	✓	×	×	×
		SCK8 (input/output)	PJ0	✓	✓	×	×
			PK0	✓	×	×	×
			PK2	✓	×	×	×
		SMOSI8 (input/output)/SSDA8 (input/output)/TXD8 (output)	PC7	✓	✓	✓	✓
			PJ2	✓	✓	×	×
			PK2	✓	×	×	×

Table 23.1 Functions Assigned to Each Multiplexed Pin (19/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Serial communications interface	SCI9	CTS9# (input)/SS9# (input)	PB4	✓	✓	✓	✓
		RTS9# (output)/CTS9# (input)/SS9# (input)	PL3	✓	×	×	×
		RTS9# (output)/SCK9 (input/output)	PB5	✓	✓	✓	✓
		RXD9 (input)/SMISO9 (input/output)/SSCL9 (input/output)	PB6	✓	✓	✓	✓
			PL1	✓	×	×	×
		SCK9 (input/output)	PL0	✓	×	×	×
		SMOSI9 (input/output)/SSDA9 (input/output)/TXD9 (output)	PB7	✓	✓	✓	✓
			PL2	✓	×	×	×
	SCI10	CTS10# (input)/RTS10# (output)/SS10# (input)	PC4	✓	✓	✓	✓
		CTS10# (input)/SCK10 (input/output)/SS10# (input)	P83	✓	✓	✓	×
		RTS10# (output)/CTS10# (input)/SS10# (input)	PM3	✓	×	×	×
		RTS10# (output)/SCK10 (input/output)	P80	✓	✓	✓	×
		RXD10 (input)/SMISO10 (input/output)/SSCL10 (input/output)	P81	✓	✓	✓	×
			P86	✓	✓	✓	×
			PC6	✓	✓	✓	✓
			PM1	✓	×	×	×
		SCK10 (input/output)	PC5	✓	✓	✓	✓
			PM0	✓	×	×	×
		SMOSI10 (input/output)/SSDA10 (input/output)/TXD10 (output)	P82	✓	✓	✓	×
			P87	✓	✓	✓	×
			PC7	✓	✓	✓	✓
			PM2	✓	×	×	×
	SCI11	CTS11# (input)/RTS11# (output)/SS11# (input)	PB4	✓	✓	✓	✓
		CTS11# (input)/SS11# (input)	P74	✓	✓	✓	×
		RTS11# (output)/CTS11# (input)/SS11# (input)	PQ3	✓	×	×	×
		RTS11# (output)/SCK11 (input/output)	P75	✓	✓	✓	×
		RXD11 (input)/SMISO11 (input/output)/SSCL11 (input/output)	P76	✓	✓	✓	×
			PB6	✓	✓	✓	✓
			PQ1	✓	×	×	×
		SCK11 (input/output)	PB5	✓	✓	✓	✓
			PQ0	✓	×	×	×
		SMOSI11 (input/output)/SSDA11 (input/output)/TXD11 (output)	P77	✓	✓	✓	×
			PB7	✓	✓	✓	✓
			PQ2	✓	×	×	×

Table 23.1 Functions Assigned to Each Multiplexed Pin (20/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Serial communications interface	SCI12	CTS12# (input)/RTS12# (output)/SS12# (input)	PE3	✓	✓	✓	✓
		RXD12 (input)/RXDX12 (input)/SMISO12 (input/output)/SSCL12 (input/output)	PE2	✓	✓	✓	✓
		SCK12 (input/output)	PE0	✓	✓	✓	✓
		SMOSI12 (input/output)/SSDA12 (input/output)/TXD12 (output)/TXDX12 (output)/SIOX12 (input/output)	PE1	✓	✓	✓	✓
I ² C bus interface	RIIC0	SCL0[FM+] (input/output)	P12	✓	✓	✓	✓
		SDA0[FM+] (input/output)	P13	✓	✓	✓	✓
	RIIC1	SCL1 (input/output)	P21	✓	✓	✓	✓
		SDA1 (input/output)	P20	✓	✓	✓	✓
	RIIC2	SCL2-DS (input/output)	P16	✓	✓	✓	✓
		SDA2-DS (input/output)	P17	✓	✓	✓	✓
USB 2.0FS host/function module	USB0	USB0_VBUS (input)	P16	✓	✓	✓	✓
		USB0_EXICEN (output)	P21	✓	✓	✓	✓
		USB0_VBUSEN (output)	P16	✓	✓	✓	✓
			P24	✓	✓	✓	✓
			P32	✓	✓	✓	✓
		USB0_OVRCURA (input)	P14	✓	✓	✓	✓
		USB0_OVRCURB (input)	P16	✓	✓	✓	✓
			P22	✓	✓	✓	✓
		USB0_ID (input)	P20	✓	✓	✓	✓
CAN module	CAN0	CRX0 (input)	P33	✓	✓	✓	✓
			PD2	✓	✓	✓	✓
		CTX0 (output)	P32	✓	✓	✓	✓
			PD1	✓	✓	✓	✓
	CAN1	CRX1-DS (input)	P15	✓	✓	✓	✓
		CRX1 (input)	P55	✓	✓	✓	✓
		CTX1 (output)	P14	✓	✓	✓	✓
			P23	✓	✓	✓	✓
			P54	✓	✓	✓	✓
	CAN2	CRX2 (input)	P67	✓	✓	✓	×
		CTX2 (output)	P66	✓	✓	✓	×

Table 23.1 Functions Assigned to Each Multiplexed Pin (21/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Serial peripheral interface	RSPI0	RSPCKA (input/output)	PA5	✓	✓	✓	✓
			PC5	✓	✓	✓	✓
			PH0	✓	x	x	x
		MOSIA (input/output)	PA6	✓	✓	✓	✓
			PC6	✓	✓	✓	✓
			PH1	✓	x	x	x
		MISOA (input/output)	PA7	✓	✓	✓	✓
			PC7	✓	✓	✓	✓
			PH2	✓	x	x	x
		SSLA0 (input/output)	PA4	✓	✓	✓	✓
			PC4	✓	✓	✓	✓
			PH3	✓	x	x	x
		SSLA1 (output)	PA0	✓	✓	✓	✓
			PC0	✓	✓	✓	✓
			PH4	✓	x	x	x
		SSLA2 (output)	PA1	✓	✓	✓	✓
			PC1	✓	✓	✓	✓
			PH5	✓	x	x	x
		SSLA3 (output)	PA2	✓	✓	✓	✓
			PC2	✓	✓	✓	✓
			PH6	✓	x	x	x
	RSPI1	RSPCKB (input/output)	P27	✓	✓	✓	✓
			PE5	✓	✓	✓	✓
			PK0	✓	x	x	x
		MOSIB (input/output)	P26	✓	✓	✓	✓
			PE6	✓	✓	✓	✓
			PK1	✓	x	x	x
		MISOB (input/output)	P30	✓	✓	✓	✓
			PE7	✓	✓	✓	✓
			PK2	✓	x	x	x
		SSLB0 (input/output)	P31	✓	✓	✓	✓
			PE4	✓	✓	✓	✓
			PK3	✓	x	x	x
		SSLB1 (output)	P50	✓	✓	✓	✓
			PE0	✓	✓	✓	✓
			PK4	✓	x	x	x
		SSLB2 (output)	P51	✓	✓	✓	✓
			PE1	✓	✓	✓	✓
			PK5	✓	x	x	x
		SSLB3 (output)	P52	✓	✓	✓	✓
			PE2	✓	✓	✓	✓
			PK6	✓	x	x	x

Table 23.1 Functions Assigned to Each Multiplexed Pin (22/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Serial peripheral interface	RSPI2	RSPCKC (input/output)	P56	✓	✓	×	×
			PD3	✓	✓	✓	✓
			PL0	✓	×	×	×
		MOSIC (input/output)	P54	✓	✓	×	×
			PD1	✓	✓	✓	✓
			PL1	✓	×	×	×
		MISOC (input/output)	P55	✓	✓	×	×
			PD2	✓	✓	✓	✓
			PL2	✓	×	×	×
		SSLC0 (input/output)	P57	✓	✓	×	×
			PD4	✓	✓	✓	✓
			PL3	✓	×	×	×
		SSLC1 (output)	PD5	✓	✓	✓	✓
			PJ0	✓	✓	×	×
			PL4	✓	×	×	×
		SSLC2 (output)	PD6	✓	✓	✓	✓
			PJ1	✓	✓	×	×
			PL5	✓	×	×	×
		SSLC3 (output)	PD7	✓	✓	✓	✓
			PJ2	✓	✓	×	×
			PL6	✓	×	×	×
Quad serial peripheral interface		QSPCLK (output)	P77	✓	✓	✓	×
			PD5	✓	✓	✓	✓
			PM0	✓	×	×	×
			PN4	✓	×	×	×
		QSSL (output)	P76	✓	✓	✓	×
			PD4	✓	✓	✓	✓
			PM1	✓	×	×	×
			PN5	✓	×	×	×
		QMO/QIO0 (input/output)	PC3	✓	✓	✓	×
			PD6	✓	✓	✓	✓
			PJ3	✓	×	×	×
			PM2	✓	×	×	×
		QMI/QIO1 (input/output)	PC4	✓	✓	✓	×
			PD7	✓	✓	✓	✓
			PJ5	✓	×	×	×
			PM3	✓	×	×	×
		QIO2 (input/output)	P00	✓	×	×	×
			P80	✓	✓	✓	×
			PD2	✓	✓	✓	✓
			PM4	✓	×	×	×

Table 23.1 Functions Assigned to Each Multiplexed Pin (23/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Quad serial peripheral interface		QIO3 (input/output)	P01	✓	×	×	×
			P81	✓	✓	✓	×
			PD3	✓	✓	✓	✓
			PM5	✓	×	×	×
Serial sound interface enhanced	SSIE	AUDIO_CLK (input)	P00	✓	✓	✓	×
			P22	✓	✓	✓	✓
	SSIE0	SSIBCK0 (input/output)	P01	✓	✓	✓	×
			P23	✓	✓	✓	✓
		SSILRCK0 (input/output)	P21	✓	✓	✓	✓
			PF5	✓	✓	✓	×
		SSIRXD0 (input)	P20	✓	✓	✓	✓
			PJ5	✓	✓	✓	×
		SSITXD0 (output)	P17	✓	✓	✓	✓
			PJ3	✓	✓	✓	✓
	SSIE1	SSIBCK1 (input/output)	P02	✓	✓	✓	×
			P24	✓	✓	✓	✓
		SSILRCK1 (input/output)	P05	✓	✓	✓	✓
			P15	✓	✓	✓	✓
		SSIDATA1 (input/output)	P03	✓	✓	✓	×
			P25	✓	✓	✓	✓
SD host interface		SDHI_CLK (output)	P21	✓	✓	✓	×
			P77	✓	✓	✓	×
			PD5	✓	✓	✓	✓
			PM0	✓	×	×	×
		SDHI_CMD (input/output)	P20	✓	✓	✓	×
			P76	✓	✓	✓	×
			PD4	✓	✓	✓	✓
			PM1	✓	×	×	×
		SDHI_CD (input)	P25	✓	✓	✓	×
			P81	✓	✓	✓	×
			PE6	✓	✓	✓	✓
			PM6	✓	×	×	×
		SDHI_WP (input)	P24	✓	✓	✓	×
			P80	✓	✓	✓	×
			PE7	✓	✓	✓	✓
			PM7	✓	×	×	×
		SDHI_D0 (input/output)	P22	✓	✓	✓	×
			PC3	✓	✓	✓	×
			PD6	✓	✓	✓	✓
			PM2	✓	×	×	×
		SDHI_D1 (input/output)	P23	✓	✓	✓	×
			PC4	✓	✓	✓	×
			PD7	✓	✓	✓	✓
			PM3	✓	×	×	×

Table 23.1 Functions Assigned to Each Multiplexed Pin (24/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
SD host interface		SDHI_D2 (input/output)	P75	✓	✓	✓	×
			P87	✓	✓	✓	×
			PD2	✓	✓	✓	✓
			PM4	✓	×	×	×
		SDHI_D3 (input/output)	P17	✓	✓	✓	×
			PC2	✓	✓	✓	×
			PD3	✓	✓	✓	✓
			PM5	✓	×	×	×
MMC host interface		MMC_RES# (output)	P75	✓	✓	✓	×
			PE7	✓	✓	✓	✓
		MMC_CLK (output)	P77	✓	✓	✓	×
			PD5	✓	✓	✓	✓
		MMC_CD (input)	PC2	✓	✓	✓	×
			PE6	✓	✓	✓	✓
		MMC_CMD (input/output)	P76	✓	✓	✓	×
			PD4	✓	✓	✓	✓
		MMC_D0 (input/output)	PC3	✓	✓	✓	×
			PD6	✓	✓	✓	✓
		MMC_D1 (input/output)	PC4	✓	✓	✓	×
			PD7	✓	✓	✓	✓
		MMC_D2 (input/output)	P80	✓	✓	✓	×
			PD2	✓	✓	✓	✓
		MMC_D3 (input/output)	P81	✓	✓	✓	×
			PD3	✓	✓	✓	✓
		MMC_D4 (input/output)	P82	✓	✓	✓	×
			PE0	✓	✓	✓	✓
		MMC_D5 (input/output)	PC5	✓	✓	✓	×
			PE1	✓	✓	✓	✓
		MMC_D6 (input/output)	PC6	✓	✓	✓	×
			PE2	✓	✓	✓	✓
		MMC_D7 (input/output)	PC7	✓	✓	✓	×
			PE3	✓	✓	✓	✓
12-bit A/D converter	S12AD	AN000 (input)*1	P40	✓	✓	✓	✓
		AN001 (input)*1	P41	✓	✓	✓	✓
		AN002 (input)*1	P42	✓	✓	✓	✓
		AN003 (input)*1	P43	✓	✓	✓	✓
		AN004 (input)*1	P44	✓	✓	✓	✓
		AN005 (input)*1	P45	✓	✓	✓	✓
		AN006 (input)*1	P46	✓	✓	✓	✓
		AN007 (input)*1	P47	✓	✓	✓	✓
		ADTRG0# (input)	P07	✓	✓	✓	✓
			P16	✓	✓	✓	✓
			P25	✓	✓	✓	✓

Table 23.1 Functions Assigned to Each Multiplexed Pin (25/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
12-bit A/D converter	S12AD1	AN100 (input)*1	PE2	✓	✓	✓	✓
		AN101 (input)*1	PE3	✓	✓	✓	✓
		AN102 (input)*1	PE4	✓	✓	✓	✓
		AN103 (input)*1	PE5	✓	✓	✓	✓
		AN104 (input)*1	PE6	✓	✓	✓	✓
		AN105 (input)*1	PE7	✓	✓	✓	✓
		AN106 (input)*1	PD6	✓	✓	✓	✓
		AN107 (input)*1	PD7	✓	✓	✓	✓
		AN108 (input)*1	PD0	✓	✓	✓	✓
		AN109 (input)*1	PD1	✓	✓	✓	✓
		AN110 (input)*1	PD2	✓	✓	✓	✓
		AN111 (input)*1	PD3	✓	✓	✓	✓
		AN112 (input)*1	PD4	✓	✓	✓	✓
		AN113 (input)*1	PD5	✓	✓	✓	✓
		AN114 (input)*1	P90	✓	✓	✓	×
		AN115 (input)*1	P91	✓	✓	✓	×
		AN116 (input)*1	P92	✓	✓	✓	×
		AN117 (input)*1	P93	✓	✓	✓	×
		AN118 (input)*1	P00	✓	✓	✓	×
		AN119 (input)*1	P01	✓	✓	✓	×
		AN120 (input)*1	P02	✓	✓	✓	×
		ANEX0 (output)*1	PE0	✓	✓	✓	✓
		ANEX1 (input)*1	PE1	✓	✓	✓	✓
		ADTRG1# (input)	P13	✓	✓	✓	✓
			P17	✓	✓	✓	✓
12-bit D/A converter		DA0 (output)*1	P03	✓	✓	✓	×
		DA1 (output)*1	P05	✓	✓	✓	✓
Parallel data capture unit		PIXCLK (input)	P24	✓	✓	✓	×
		VSYN (input)	P32	✓	✓	✓	×
		HSYN (input)	P25	✓	✓	✓	×
		PIXD0 (input)	P15	✓	✓	✓	×
		PIXD1 (input)	P86	✓	✓	✓	×
		PIXD2 (input)	P87	✓	✓	✓	×
		PIXD3 (input)	P17	✓	✓	✓	×
		PIXD4 (input)	P20	✓	✓	✓	×
		PIXD5 (input)	P21	✓	✓	✓	×
		PIXD6 (input)	P22	✓	✓	✓	×
		PIXD7 (input)	P23	✓	✓	✓	×
		PCKO (output)	P33	✓	✓	✓	×
Clock generation circuit		CLKOUT (output)	P25	✓	✓	✓	✓
			PH6	✓	×	×	×
		CLKOUT25M (output)	P56	✓	✓	✓	×
			PH7	✓	×	×	×
			PJ2	✓	✓	×	×

Table 23.1 Functions Assigned to Each Multiplexed Pin (26/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Clock frequency accuracy measurement circuit		CACREF (input)	PA0	✓	✓	✓	✓
			PC7	✓	✓	✓	✓
Graphic-LCD controller		LCD_EXTCLK (input)	P73	✓	✓	x	x
			PD0	✓	✓	✓	✓
		LCD_CLK (output)	P14	✓	✓	x	x
			PB5	✓	✓	✓	✓
		LCD_TCON0 (output)	P13	✓	✓	x	x
			PB4	✓	✓	✓	✓
		LCD_TCON1 (output)	PB3	✓	✓	✓	✓
			P12	✓	✓	x	x
		LCD_TCON2 (output)	PB2	✓	✓	✓	✓
			PJ2	✓	✓	x	x
		LCD_TCON3 (output)	PB1	✓	✓	✓	✓
			PJ1	✓	✓	x	x
		LCD_DATA0 (output)	PB0	✓	✓	✓	✓
			PJ0	✓	✓	x	x
		LCD_DATA1 (output)	P85	✓	✓	x	x
			PA7	✓	✓	✓	✓
		LCD_DATA2 (output)	P84	✓	✓	x	x
			PA6	✓	✓	✓	✓
		LCD_DATA3 (output)	P57	✓	✓	x	x
			PA5	✓	✓	✓	✓
		LCD_DATA4 (output)	P56	✓	✓	x	x
			PA4	✓	✓	✓	✓
		LCD_DATA5 (output)	P55	✓	✓	x	x
			PA3	✓	✓	✓	✓
		LCD_DATA6 (output)	P54	✓	✓	x	x
			PA2	✓	✓	✓	✓
		LCD_DATA7 (output)	P11	✓	✓	x	x
			PA1	✓	✓	✓	✓
		LCD_DATA8 (output)	P83	✓	✓	x	x
			PA0	✓	✓	✓	✓
		LCD_DATA9 (output)	PC7	✓	✓	x	x
			PE7	✓	✓	✓	✓
		LCD_DATA10 (output)	PC6	✓	✓	x	x
			PE6	✓	✓	✓	✓
		LCD_DATA11 (output)	PC5	✓	✓	x	x
			PE5	✓	✓	✓	✓
		LCD_DATA12 (output)	P82	✓	✓	x	x
			PE4	✓	✓	✓	✓
		LCD_DATA13 (output)	P81	✓	✓	x	x
			PE3	✓	✓	✓	✓
		LCD_DATA14 (output)	P80	✓	✓	x	x
			PE2	✓	✓	✓	✓

Table 23.1 Functions Assigned to Each Multiplexed Pin (27/27)

Module/Function	Channel	Pin Functions	Allocation Port	Package			
				224-pin	176-pin	145-pin 144-pin	100-pin
Graphic-LCD controller		LCD_DATA15 (output)	PC4	✓	✓	×	×
			PE1	✓	✓	✓	✓
		LCD_DATA16 (output)	PC3	✓	✓	×	×
			PE0	✓	✓	✓	✓
		LCD_DATA17 (output)	P77	✓	✓	×	×
			PD7	✓	✓	✓	✓
		LCD_DATA18 (output)	P76	✓	✓	×	×
			PD6	✓	✓	✓	✓
		LCD_DATA19 (output)	PC2	✓	✓	×	×
			PD5	✓	✓	✓	✓
		LCD_DATA20 (output)	P75	✓	✓	×	×
			PD4	✓	✓	✓	✓
		LCD_DATA21 (output)	P74	✓	✓	×	×
			PD3	✓	✓	✓	✓
		LCD_DATA22 (output)	PC1	✓	✓	×	×
			PD2	✓	✓	✓	✓
		LCD_DATA23 (output)	P72	✓	✓	×	×
			PD1	✓	✓	✓	✓

Note 1. To use this pin function, set the corresponding pin as general input (set the PORTm.PDR.Bn and PORTm.PMR.Bn bits to 0).

23.2 Register Descriptions

The registers and bits of unsupported pins, depending on the package, are reserved. The write value to the reserved bits is the value after a reset.

23.2.1 Write-Protect Register (PWPR)

Address(es): 0008 C11Fh

	b7	b6	b5	b4	b3	b2	b1	b0
	B0WI	PFSWE	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b5 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b6	PFSWE	PFS Register Write Enable	0: Writing to the PFS register is disabled 1: Writing to the PFS register is enabled	R/W
b7	B0WI	PFSWE Bit Write Disable	0: Writing to the PFSWE bit is enabled 1: Writing to the PFSWE bit is disabled	R/W

PFSWE Bit (PFS Register Write Enable)

Writing to PmnPFS register is enabled only when the PFSWE bit is set to 1.

To set the PFSWE bit to 1, write 1 to the PFSWE bit after writing 0 to the B0WI bit.

B0WI Bit (PFSWE Bit Write Disable)

Writing to the PFSWE bit is enabled only when the B0WI bit is set to 0.

23.2.2 P0n Pin Function Control Register (P0nPFS) (n = 0 to 3, 5, 7)

Address(es): P00PFS 0008 C140h, P01PFS 0008 C141h, P02PFS 0008 C142h, P03PFS 0008 C143h,
P05PFS 0008 C145h, P07PFS 0008 C147h



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.2 and Table 23.3.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin P00: IRQ8 (224/176/145/144 pin) P01: IRQ9 (224/176/145/144 pin) P02: IRQ10 (224/176/145/144 pin) P03: IRQ11 (224/176/145/144 pin) P05: IRQ13 (224/176/145/144/100 pin) P07: IRQ15 (224/176/145/144/100 pin)	R/W
b7	ASEL	Analog Function Select	0: Used other than as analog pin. 1: Used as analog pin. P00: AN118 (224/176/145/144 pin) P01: AN119 (224/176/145/144 pin) P02: AN120 (224/176/145/144 pin) P03: DA0 (224/176/145/144 pin) P05: DA1 (224/176/145/144/100 pin)	R/W

The port mn pin function control register (PmnPFS) selects the pin function. Bits PSEL[5:0] select the peripheral function which is assigned to bits.

The ISEL bit is set when a pin is used as an IRQ input pin. This setting can be used with the combination of the peripheral function, though IRQn (external pin interrupt) of the same number should not be enabled by two or more pins.

The ASEL bit is set when a pin is used as an analog pin. When the pin is set as an analog pin by the ASEL bit, select the general I/O port by the port mode register (PORTm.PMR) and specify input by the port direction register (PORTm.PDR). The pin state cannot be read at this point, since the PmnPFS register is protected by the write-protect register (PWPR). Modify the register after releasing the protection.

The ISEL bit to which IRQn is not specified is reserved. The ASEL bit to which analog input/output is not specified is reserved.

Table 23.2 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-Pin LFQFP, 145-Pin TFLGA

PSEL[5:0] Settings	Pin					
	P00	P01	P02	P03	P05	P07
000000b (initial value)	Hi-Z					
000101b	TMRI0	TMCIO	TMC11	—	—	—
001001b	—	—	—	—	—	ADTRG0#
001010b	TXD6 SMOSI6 SSDA6	RXD6 SMISO6 SSCL6	SCK6	—	—	—
010111b	AUDIO_CLK	SSIBCK0	SSIBCK1	SSIDATA1	SSILRCK1	—
011011b	QIO2-C*1	QIO3-C*1	—	—	—	—

—: Do not specify this value.

Note 1. This setting is not supported by 176-/145-/144-pin products.

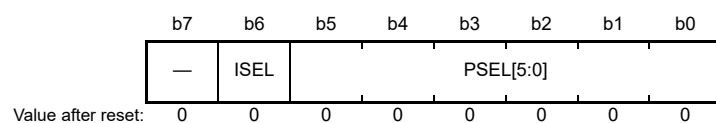
Table 23.3 Register Settings for Input/Output Pin Function in 100-Pin LFQFP

PSEL[5:0] Settings	Pin	
	P05	P07
000000b (initial value)	Hi-Z	
001001b	—	ADTRG0#
010111b	SSILRCK1	—

—: Do not specify this value.

23.2.3 P1n Pin Function Control Register (P1nPFS) (n = 0 to 7)

Address(es): P10PFS 0008 C148h, P11PFS 0008 C149h, P12PFS 0008 C14Ah, P13PFS 0008 C14Bh, P14PFS 0008 C14Ch, P15PFS 0008 C14Dh, P16PFS 0008 C14Eh, P17PFS 0008 C14Fh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.4 and Table 23.5.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin P10: IRQ0 (224/176 pin) P11: IRQ1 (224/176 pin) P12: IRQ2 (224/176/145/144/100 pin) P13: IRQ3 (224/176/145/144/100 pin) P14: IRQ4 (224/176/145/144/100 pin) P15: IRQ5 (224/176/145/144/100 pin) P16: IRQ6 (224/176/145/144/100 pin) P17: IRQ7 (224/176/145/144/100 pin)	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Table 23.4 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-Pin LFQFP

PSEL[5:0] Settings	Pin							
	P10	P11	P12	P13	P14	P15	P16	P17
000000b (initial value)	Hi-Z							
000001b	MTIC5W	MTIC5V	MTIC5U	MTIOC0B	MTIOC3A	MTIOC0B	MTIOC3C	MTIOC3A
000010b	—	—	—	—	MTCLKA	MTCLKB	MTIOC3D	MTIOC3B
000011b	—	—	—	TIOCA5	TIOCB5	TIOCB2	TIOCB1	TIOCB0
000100b	—	—	—	—	TCLKA	TCLKB	TCLKC	TCLKD
000101b	TMRI3	TMCI3	TMCI1	TMO3	TMRI2	TMCI2	TMO2	TMO1
000110b	—	—	—	PO13	PO15	PO13	PO14	PO15
000111b	—	—	—	—	—	—	RTCOUT	POE8#
001000b	—	—	—	—	—	—	—	MTIOC4B
001001b	—	—	—	ADTRG1#	—	—	ADTRG0#	ADTRG1#
001010b	—	SCK2	RXD2 SMISO2 SSCL2	TXD2 SMOSI2 SSDA2	—	RXD1 SMISO1 SSCL1	TXD1 SMOSI1 SSDA1	SCK1
001011b	—	—	—	—	CTS1# RTS1# SS1#	SCK3	RXD3 SMISO3 SSCL3	TXD3 SMOSI3 SSDA3
001111b	—	—	SCL0[FM+]	SDA0[FM+]	—	—	SCL2-DS	SDA2-DS
010000b	—	—	—	—	CTX1	CRX1-DS	—	—
010001b	—	EPLSOUT1	—	—	—	—	USB0_VBUS	EPLSOUT0
010010b	—	—	—	—	USB0_OVRC URA	—	USB0_VBUS EN	—
010011b	—	—	—	—	—	—	USB0_OVRC URB	—
010111b	—	—	—	—	—	SSILRCK1	—	SSITXD0
011010b	—	—	—	—	—	—	—	SDHI_D3-C
011100b	—	—	—	—	—	PIXD0	—	PIXD3
011110b	—	—	GTADSM0	GTADSM1	GTETRGD	GTETRGA	—	GTIOC0B
100101b	—	LCD_DATA7- A	LCD_TCON1 -A	LCD_TCON0 -A	LCD_CLK-A	—	—	—

—: Do not specify this value.

Table 23.5 Register Settings for Input/Output Pin Function in 145-Pin TFLGA, 144-/100-Pin LFQFP

PSEL[5:0] Settings	Pin					
	P12	P13	P14	P15	P16	P17
000000b (initial value)	Hi-Z					
000001b	—	MTIOC0B	MTIOC3A	MTIOC0B	MTIOC3C	MTIOC3A
000010b	—	—	MTCLKA	MTCLKB	MTIOC3D	MTIOC3B
000011b	—	TIOCA5	TIOCB5	TIOCB2	TIOCB1	TIOCB0
000100b	—	—	TCLKA	TCLKB	TCLKC	TCLKD
000101b	TMCI1	TMO3	TMRI2	TMCI2	TMO2	TMO1
000110b	—	PO13	PO15	PO13	PO14	PO15
000111b	—	—	—	—	RTCOUT	POE8#
001000b	—	—	—	—	—	MTIOC4B
001001b	—	ADTRG1#	—	—	ADTRG0#	ADTRG1#
001010b	RXD2 SMISO2 SSCL2	TXD2 SMOSI2 SSDA2	—	RXD1 SMISO1 SSCL1	TXD1 SMOSI1 SSDA1	SCK1
001011b	—	—	CTS1# RTS1# SS1#	SCK3	RXD3 SMISO3 SSCL3	TXD3 SMOSI3 SSDA3
001111b	SCL0[FM+]	SDA0[FM+]	—	—	SCL2-DS	SDA2-DS
010000b	—	—	CTX1	CRX1-DS	—	—
010001b	—	—	—	—	USB0_VBUS	EPLSOUT0
010010b	—	—	USB0_OVRC URA	—	USB0_VBUS EN	—
010011b	—	—	—	—	USB0_OVRC URB	—
010111b	—	—	—	SSILRCK1	—	SSITXD0
011010b	—	—	—	—	—	SDHI_D3-C*1
011100b	—	—	—	PIXD0*1	—	PIXD3*1
011110b	GTADSM0	GTADSM1	GTETRGD	GTETRGA	—	GTIOC0B

—: Do not specify this value.

Note 1. This setting is not supported by 100-pin products.

23.2.4 P2n Pin Function Control Register (P2nPFS) (n = 0 to 7)

Address(es): P20PFS 0008 C150h, P21PFS 0008 C151h, P22PFS 0008 C152h, P23PFS 0008 C153h, P24PFS 0008 C154h, P25PFS 0008 C155h, P26PFS 0008 C156h, P27PFS 0008 C157h



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.6.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin P20: IRQ8 (224/176/145/144/100 pin) P21: IRQ9 (224/176/145/144/100 pin)	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Table 23.6 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-/100-Pin LQFP, 145-Pin TFLGA

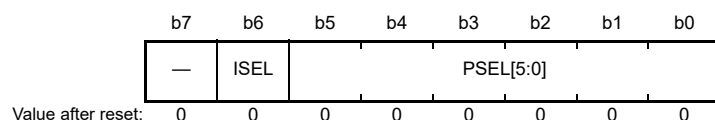
PSEL[5:0] Settings	Pin							
	P20	P21	P22	P23	P24	P25	P26	P27
000000b (initial value)	Hi-Z							
000001b	MTIOC1A	MTIOC1B	MTIOC3B	MTIOC3D	MTIOC4A	MTIOC4C	MTIOC2A	MTIOC2B
000010b	—	—	MTCLKC	MTCLKD	MTCLKA	MTCLKB	—	—
000011b	TIOCB3	TIOCA3	TIOCC3	TIOCD3	TIOCB4	TIOCA4	—	—
000101b	TMRI0	TMCIO	TMO0	—	TMRI1	—	TMO1	TMCIO
000110b	PO0	PO1	PO2	PO3	PO4	PO5	PO6	PO7
001000b	—	MTIOC4A	—	—	—	—	—	—
001001b	—	—	—	—	—	ADTRG0#	—	—
001010b	TXD0 SMOSI0 SSDA0	RXD0 SMISO0 SSCL0	SCK0	TXD3 SMOSI3 SSDA3	SCK3	RXD3 SMISO3 SSCL3	TXD1 SMOSI1 SSDA1	SCK1
001011b	—	—	—	CTS0# RTS0# SS0#	—	—	CTS3# RTS3# SS3#	—
001101b	—	—	—	—	—	—	MOSIB-A	RSPCKB-A
001111b	SDA1	SCL1	—	—	—	—	—	—
010000b	—	—	—	CTX1	—	—	—	—
010011b	USB0_ID	USB0_EXICEN	USB0_OVRCURB	—	USB0_VBUSEN	—	—	—
010100b	—	—	—	—	—	—	ET1_EXOUT*1	ET1_WOL*1
010111b	SSIRXD0	SSILRCK0	AUDIO_CLK	SSIBCK0	SSIBCK1	SSIDATA1	—	—
011000b	—	—	EDREQ0	EDACK0	EDREQ1	EDACK1	—	—
011010b	SDHI_CMD-C*1	SDHI_CLK-C*1	SDHI_D0-C*1	SDHI_D1-C*1	SDHI_WP*1	SDHI_CD*1	—	—
011100b	PIXD4*1	PIXD5*1	PIXD6*1	PIXD7*1	PIXCLK*1	HSYNC*1	—	—
011110b	—	GTIOC2A	GTIOC1A	GTIOC0A	—	—	—	—
101010b	—	—	—	—	—	CLKOUT	—	—

—: Do not specify this value.

Note 1. This setting is not supported by 100-pin products.

23.2.5 P3n Pin Function Control Register (P3nPFS) (n = 0 to 4)

Address(es): P30PFS 0008 C158h, P31PFS 0008 C159h, P32PFS 0008 C15Ah, P33PFS 0008 C15Bh, P34PFS 0008 C15Ch



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.7.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin P30: IRQ0-DS (224/176/145/144/100 pin) P31: IRQ1-DS (224/176/145/144/100 pin) P32: IRQ2-DS (224/176/145/144/100 pin) P33: IRQ3-DS (224/176/145/144/100 pin) P34: IRQ4 (224/176/145/144/100 pin)	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Table 23.7 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-/100-Pin LFQFP, 145-Pin TFLGA

PSEL[5:0] Settings	Pin				
	P30	P31	P32	P33	P34
000000b (initial value)	Hi-Z				
000001b	MTIOC4B	MTIOC4D	MTIOC0C	MTIOC0D	MTIOC0A
000011b	—	—	TIOCC0	TIOCD0	—
000101b	TMRI3	TMCI2	TMO3	TMRI3	TMCI3
000110b	PO8	PO9	PO10	PO11	PO12
000111b	POE8#	—	RTCOUT	—	POE10#
001000b	—	—	POE0#	POE4#	—
001010b	RXD1 SMISO1 SSCL1	—	TXD6 SMOSI6 SSDA6	RXD6 SMISO6 SSCL6	SCK6
001011b	—	CTS1# RTS1# SS1#	TXD0 SMOSI0 SSDA0	RXD0 SMISO0 SSCL0	SCK0
001101b	MISOB-A	SSLB0-A	—	—	—
010000b	—	—	CTX0	CRX0	—
010001b	—	—	—	—	ET0_LINKST A
010011b	—	—	USB0_VBUS EN	—	—
010100b	ET1_MDIO*1	ET1_MDC*1	—	—	—
011000b	—	—	—	EDREQ1	—
011100b	—	—	VSYNC*1	PCKO*1	—
100001b	—	—	POE10#	POE11#	—
101000b	PMGI1_MDI O*1	PMGI1_MDC *1	—	—	—

—: Do not specify this value.

Note 1. This setting is not supported by 100-pin products.

23.2.6 P4n Pin Function Control Register (P4nPFS) (n = 0 to 7)

Address(es): P40PFS 0008 C160h, P41PFS 0008 C161h, P42PFS 0008 C162h, P43PFS 0008 C163h,
P44PFS 0008 C164h, P45PFS 0008 C165h, P46PFS 0008 C166h, P47PFS 0008 C167h

b7	b6	b5	b4	b3	b2	b1	b0
ASEL	ISEL	—	—	—	—	—	—

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b5 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin P40: IRQ8-DS (224/176/145/144/100 pin) P41: IRQ9-DS (224/176/145/144/100 pin) P42: IRQ10-DS (224/176/145/144/100 pin) P43: IRQ11-DS (224/176/145/144/100 pin) P44: IRQ12-DS (224/176/145/144/100 pin) P45: IRQ13-DS (224/176/145/144/100 pin) P46: IRQ14-DS (224/176/145/144/100 pin) P47: IRQ15-DS (224/176/145/144/100 pin)	R/W
b7	ASEL	Analog Input Function Select	0: Used other than as analog pin. 1: Used as analog pin. P40: AN000 (224/176/145/144/100 pin) P41: AN001 (224/176/145/144/100 pin) P42: AN002 (224/176/145/144/100 pin) P43: AN003 (224/176/145/144/100 pin) P44: AN004 (224/176/145/144/100 pin) P45: AN005 (224/176/145/144/100 pin) P46: AN006 (224/176/145/144/100 pin) P47: AN007 (224/176/145/144/100 pin)	R/W

23.2.7 P5n Pin Function Control Register (P5nPFS) (n = 0 to 2, 4 to 7)

Address(es): P50PFS 0008 C168h, P51PFS 0008 C169h, P52PFS 0008 C16Ah, P54PFS 0008 C16Ch,
P55PFS 0008 C16Dh, P56PFS 0008 C16Eh, P57PFS 0008 C16Fh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.8 to Table 23.10.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin P55: IRQ10 (224/176/145/144/100 pin)	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Table 23.8 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-Pin LFQFP

PSEL[5:0] Settings	Pin						
	P50	P51	P52	P54	P55	P56	P57
000000b (initial value)	Hi-Z						
000001b	—	—	—	MTIOC4B	MTIOC4D	MTIOC3C	—
000011b	—	—	—	—	—	TIOCA1	—
000101b	—	—	—	TMC11	TMO3	—	—
001010b	TXD2 SMOSI2 SSDA2	SCK2	RXD2 SMISO2 SSCL2	—	TXD7 SMOSI7 SSDA7	SCK7	RXD7 SMISO7 SSCL7
001011b	—	—	—	CTS2# RTS2# SS2#	—	—	—
001101b	SSLB1-A	SSLB2-A	SSLB3-A	MOSIC-B	MISOC-B	RSPCKC-B	SSLC0-B
010000b	—	—	—	CTX1	CRX1	—	—
010001b	—	—	—	ET0_LINKST A	ET0_EXOUT	—	—
011000b	—	—	—	EDACK0	EDREQ0	EDACK1	—
100101b	—	—	—	LCD_DATA6- A	LCD_DATA5- A	LCD_DATA4- A	LCD_DATA3- A
101010b	—	—	—	—	—	CLKOUT25M	—

—: Do not specify this value.

Table 23.9 Register Settings for Input/Output Pin Function in 145-Pin TFLGA, 144-Pin LFQFP

PSEL[5:0] Settings	Pin					
	P50	P51	P52	P54	P55	P56
000000b (initial value)	Hi-Z					
000001b	—	—	—	MTIOC4B	MTIOC4D	MTIOC3C
000011b	—	—	—	—	—	TIOCA1
000101b	—	—	—	TMCI1	TMO3	—
001010b	TXD2 SMOSI2 SSDA2	SCK2	RXD2 SMISO2 SSCL2	—	TXD7 SMOSI7 SSDA7	SCK7
001011b	—	—	—	CTS2# RTS2# SS2#	—	—
001101b	SSLB1-A	SSLB2-A	SSLB3-A	—	—	—
010000b	—	—	—	CTX1	CRX1	—
010001b	—	—	—	ET0_LINKST A	ET0_EXOUT	—
011000b	—	—	—	EDACK0	EDREQ0	EDACK1
101010b	—	—	—	—	—	CLKOUT25M

—: Do not specify this value.

Table 23.10 Register Settings for Input/Output Pin Function in 100-Pin LFQFP

PSEL[5:0] Settings	Pin				
	P50	P51	P52	P54	P55
000000b (initial value)	Hi-Z				
000001b	—	—	—	MTIOC4B	MTIOC4D
000101b	—	—	—	TMCI1	TMO3
001010b	TXD2 SMOSI2 SSDA2	SCK2	RXD2 SMISO2 SSCL2	—	—
001011b	—	—	—	CTS2# RTS2# SS2#	—
001101b	SSLB1-A	SSLB2-A	SSLB3-A	—	—
010000b	—	—	—	CTX1	CRX1
010001b	—	—	—	ET0_LINKST A	ET0_EXOUT
011000b	—	—	—	EDACK0	EDREQ0

—: Do not specify this value.

23.2.8 P6n Pin Function Control Register (P6nPFS) (n = 0 to 4, 6, 7)

Address(es): P60PFS 0008 C170h, P61PFS 0008 C171h, P62PFS 0008 C172h, P63PFS 0008 C173h,
P64PFS 0008 C174h, P66PFS 0008 C176h, P67PFS 0008 C177h



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.11.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin P67: IRQ15 (224/176/145/144 pin)	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Table 23.11 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-Pin LFQFP, 145-Pin TFLGA

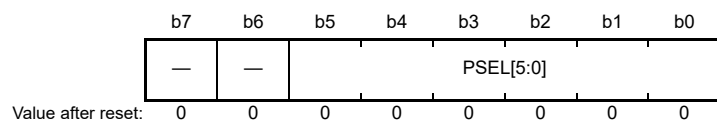
PSEL[5:0] Settings	Pin						
	P60	P61	P62	P63	P64	P66	P67
000000b (initial value)	Hi-Z						
001000b	—	—	—	—	—	MTIOC7D	MTIOC7C
010000b	—	—	—	—	—	CTX2	CRX2
010001b	—	—	—	—	—	—	EPLSOUT1
010100b	ET1_TX_EN *1	ET1_ERXD1 *1	ET1_ERXD0 *1	ET1_ETXD1 *1	ET1_ETXD0 *1	—	—
010101b	RMII1_TXD_EN	RMII1_RXD1	RMII1_RXD0	RMII1_TXD1	RMII1_TXD0	—	—
011110b	—	—	—	—	—	GTIOC2B	GTIOC1B

—: Do not specify this value.

Note 1. This setting is not supported by 145-/144-pin products.

23.2.9 P7n Pin Function Control Register (P7nPFS) (n = 1 to 7)

Address(es): P71PFS 0008 C179h, P72PFS 0008 C17Ah, P73PFS 0008 C17Bh,
P74PFS 0008 C17Ch, P75PFS 0008 C17Dh, P76PFS 0008 C17Eh, P77PFS 0008 C17Fh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.12.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Table 23.12 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-Pin LQFP, 145-Pin TFLGA

PSEL[5:0] Settings	Pin						
	P71	P72	P73	P74	P75	P76	P77
000000b (initial value)	Hi-Z						
000110b	—	—	PO16	PO19	PO20	PO22	PO23
001010b	—	—	—	—	SCK11	SMISO11 SSCL11 RXD11	SMOSI11 SSDA11 TXD11
001011b	—	—	—	SS11# CTS11#	RTS11#	—	—
010001b	ET0_MDIO	ET0_MDC	ET0_WOL	ET0_ERXD1	ET0_ERXD0	ET0_RX_CLK	ET0_RX_ER
010010b	—	—	—	RMII0_RXD1	RMII0_RXD0	REF50CK0	RMII0_RX_ER
011001b	—	—	—	—	MMC_RES#-A	MMC_CMD-A	MMC_CLK-A
011010b	—	—	—	—	SDHI_D2-A	SDHI_CMD-A	SDHI_CLK-A
011011b	—	—	—	—	—	QSSL-A	QSPCLK-A
100101b	—	LCD_DATA2 3-A* [†]	LCD_EXTCLK-A* [†]	LCD_DATA2 1-A* [†]	LCD_DATA2 0-A* [†]	LCD_DATA1 8-A* [†]	LCD_DATA1 7-A* [†]
101000b	PMGIO_MDI O	PMGIO_MDC	—	—	—	—	—

—: Do not specify this value.

Note 1. This setting is not supported by 145-/144-pin products.

23.2.10 P8n Pin Function Control Register (P8nPFS) (n = 0 to 7)

Address(es): P80PFS 0008 C180h, P81PFS 0008 C181h, P82PFS 0008 C182h, P83PFS 0008 C183h,
P84PFS 0008 C184h, P85PFS 0008 C185h, P86PFS 0008 C186h, P87PFS 0008 C187h



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.13 and Table 23.14.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Table 23.13 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-Pin LQFP

PSEL[5:0] Settings	Pin							
	P80	P81	P82	P83	P84	P85	P86	P87
000000b (initial value)	Hi-Z							
000001b	MTIOC3B	MTIOC3D	MTIOC4A	MTIOC4C	MTIOC6D	MTIOC6C	—	—
000011b	—	—	—	—	—	TIOCC0	TIOCA0	TIOCA2
000110b	PO26	PO27	PO28	—	—	—	—	—
001000b	—	—	—	—	—	—	MTIOC4D	MTIOC4C
001010b	SCK10	SMISO10 SSCL10 RXD10	SMOSI10 SSDA10 TXD10	SCK10	—	—	SMISO10 SSCL10 RXD10	SMOSI10 SSDA10 TXD10
001011b	RTS10#	—	—	SS10# CTS10#	—	—	—	—
010001b	ET0_TX_EN	ET0_ETXD0	ET0_ETXD1	ET0_CRS	—	—	—	EPLSOUT1
010010b	RMII0_TXD_EN	RMII0_TXD0	RMII0_TXD1	RMII0_CRS_DV	—	—	—	—
010100b	—	—	—	—	ET1_LINKST A	—	—	—
011000b	EDREQ0	EDACK0	EDREQ1	EDACK1	—	—	—	—
011001b	MMC_D2-A	MMC_D3-A	MMC_D4-A	—	—	—	—	—
011010b	SDHI_WP	SDHI_CD	—	—	—	—	—	SDHI_D2-C
011011b	QIO2-A	QIO3-A	—	—	—	—	—	—
011100b	—	—	—	—	—	—	PIXD1	PIXD2
011110b	—	GTIOC0B	GTIOC2A	GTIOC0A	—	—	GTIOC2B	GTIOC1B
100101b	LCD_DATA1 4-A	LCD_DATA1 3-A	LCD_DATA1 2-A	LCD_DATA8-A	LCD_DATA2-A	LCD_DATA1-A	—	—

—: Do not specify this value.

Table 23.14 Register Settings for Input/Output Pin Function in 145-Pin TFLGA, 144-Pin LFQFP

PSEL[5:0] Settings	Pin					
	P80	P81	P82	P83	P86	P87
000000b (initial value)	Hi-Z					
000001b	MTIOC3B	MTIOC3D	MTIOC4A	MTIOC4C	—	—
000011b	—	—	—	—	TIOCA0	TIOCA2
000110b	PO26	PO27	PO28	—	—	—
001000b	—	—	—	—	MTIOC4D	MTIOC4C
001010b	SCK10	SMISO10 SSCL10 RXD10	SMOSI10 SSDA10 TXD10	SCK10	SMISO10 SSCL10 RXD10	SMOSI10 SSDA10 TXD10
001011b	RTS10#	—	—	SS10# CTS10#	—	—
010001b	ET0_TX_EN	ET0_ETXD0	ET0_ETXD1	ET0_CRS	—	EPLSOUT1
010010b	RMII0_TXD_EN	RMII0_TXD0	RMII0_TXD1	RMII0_CRS_DV	—	—
011000b	EDREQ0	EDACK0	EDREQ1	EDACK1	—	—
011001b	MMC_D2-A	MMC_D3-A	MMC_D4-A	—	—	—
011010b	SDHI_WP	SDHI_CD	—	—	—	SDHI_D2-C
011011b	QIO2-A	QIO3-A	—	—	—	—
011100b	—	—	—	—	PIXD1	PIXD2
011110b	—	GTIOC0B	GTIOC2A	GTIOC0A	GTIOC2B	GTIOC1B

—: Do not specify this value.

23.2.11 P9n Pin Function Control Register (P9nPFS) (n = 0 to 7)

Address(es): P90PFS 0008 C188h, P91PFS 0008 C189h, P92PFS 0008 C18Ah, P93PFS 0008 C18Bh, P94PFS 0008 C18Ch, P95PFS 0008 C18Dh, P96PFS 0008 C18Eh, P97PFS 0008 C18Fh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.15 and Table 23.16.	R/W
b6	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7	ASEL	Analog Input Function Select	0: Used other than as analog pin. 1: Used as analog pin. P90: AN114 (224/176/145/144 pin) P91: AN115 (224/176/145/144 pin) P92: AN116 (224/176/145/144 pin) P93: AN117 (224/176/145/144 pin)	R/W

Table 23.15 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-Pin LFQFP

PSEL[5:0] Settings	Pin							
	P90	P91	P92	P93	P94	P95	P96	P97
000000b (initial value)	Hi-Z							
001000b	—	—	POE4#	POE0#	—	—	—	—
001010b	TXD7 SMOSI7 SSDA7	SCK7	RXD7 SMISO7 SSCL7	—	—	—	—	—
001011b	—	—	—	CTS7# RTS7# SS7#	—	—	—	—
010100b	ET1_RX_DV	ET1_COL	ET1_CRS	ET1_LINKST A	ET1_ERXD0	ET1_ERXD1	ET1_ERXD2	ET1_ERXD3
010101b	—	—	RMII1_CRS_ DV	—	RMII1_RXD0	RMII1_RXD1	—	—

—: Do not specify this value.

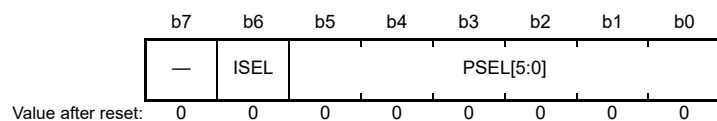
Table 23.16 Register Settings for Input/Output Pin Function in 145-Pin TFLGA, 144-Pin LFQFP

PSEL[5:0] Settings	Pin			
	P90	P91	P92	P93
000000b (initial value)	Hi-Z			
001000b	—	—	POE4#	POE0#
001010b	TXD7 SMOSI7 SSDA7	SCK7	RXD7 SMISO7 SSCL7	—
001011b	—	—	—	CTS7# RTS7# SS7#
010100b	—	—	—	ET1_LINKST A
010101b	—	—	RMII1_CRS_ DV	—

—: Do not specify this value.

23.2.12 PAn Pin Function Control Register (PAnPFS) (n = 0 to 7)

Address(es): PA0PFS 0008 C190h, PA1PFS 0008 C191h, PA2PFS 0008 C192h, PA3PFS 0008 C193h,
PA4PFS 0008 C194h, PA5PFS 0008 C195h, PA6PFS 0008 C196h, PA7PFS 0008 C197h



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.17.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin PA1: IRQ11 (224/176/145/144/100 pin) PA3: IRQ6-DS (224/176/145/144/100 pin) PA4: IRQ5-DS (224/176/145/144/100 pin)	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

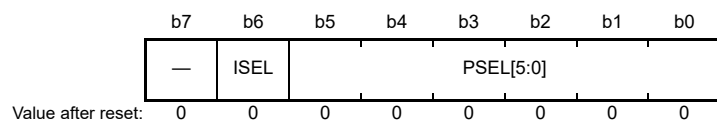
Table 23.17 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-/100-Pin LQFP, 145-Pin TFLGA

PSEL[5:0] Settings	Pin							
	PA0	PA1	PA2	PA3	PA4	PA5	PA6	PA7
000000b (initial value)	Hi-Z							
000001b	MTIOC4A	MTIOC0B	—	MTIOC0D	MTIC5U	—	MTIC5V	—
000010b	—	MTCLKC	—	MTCLKD	MTCLKA	—	MTCLKB	—
000011b	TIOCA0	TIOCB0	—	TIOCDO	TIOCA1	TIOCB1	TIOCA2	TIOCB2
000100b	—	—	—	TCLKB	—	—	—	—
000101b	—	—	—	—	TMRI0	—	TMCI3	—
000110b	PO16	PO17	PO18	PO19	PO20	PO21	PO22	PO23
000111b	CACREF	—	—	—	—	—	POE10#	—
001000b	MTIOC6D	MTIOC7B	MTIOC7A	—	—	MTIOC6B	—	—
001010b	—	SCK5	RXD5 SMISO5 SSCL5	RXD5 SMISO5 SSCL5	TXD5 SMOSI5 SSDA5	—	—	—
001011b	—	—	—	—	—	—	CTS5# RTS5# SS5#	—
001101b	SSLA1-B	SSLA2-B	SSLA3-B	—	SSLA0-B	RSPCKA-B	MOSIA-B	MISOA-B
010001b	ET0_TX_EN	ET0_WOL	—	ET0_MDIO	ET0_MDC	ET0_LINKST A	ET0_EXOUT	ET0_WOL
010010b	RMII0_TXD_ EN	—	—	—	—	—	—	—
011110b	GTIOC0B	GTIOC2A	GTIOC1A	—	—	GTIOC0A	GTETRGB	—
100101b	LCD_DATA8- B	LCD_DATA7- B	LCD_DATA6- B	LCD_DATA5- B	LCD_DATA4- B	LCD_DATA3- B	LCD_DATA2- B	LCD_DATA1- B
101000b	—	—	—	PMGI0_MDI O	PMGI0_MDC	—	—	—

—: Do not specify this value.

23.2.13 PBn Pin Function Control Register (PBnPFS) (n = 0 to 7)

Address(es): PB0PFS 0008 C198h, PB1PFS 0008 C199h, PB2PFS 0008 C19Ah, PB3PFS 0008 C19Bh, PB4PFS 0008 C19Ch, PB5PFS 0008 C19Dh, PB6PFS 0008 C19Eh, PB7PFS 0008 C19Fh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.18.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin PB0: IRQ12 (224/176/145/144/100 pin) PB1: IRQ4-DS (224/176/145/144/100 pin)	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Table 23.18 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-/100-Pin LQFP, 145-Pin TFLGA

PSEL[5:0] Settings	Pin							
	PB0	PB1	PB2	PB3	PB4	PB5	PB6	PB7
000000b (initial value)	Hi-Z							
000001b	MTIC5W	MTIOC0C	—	MTIOC0A	—	MTIOC2A	MTIOC3D	MTIOC3B
000010b	—	MTIOC4C	—	MTIOC4A	—	MTIOC1B	—	—
000011b	TIOCA3	TIOCB3	TIOCC3	TIOCD3	TIOCA4	TIOCB4	TIOCA5	TIOCB5
000100b	—	—	TCLKC	TCLKD	—	—	—	—
000101b	—	TMCIO	—	TMO0	—	TMRI1	—	—
000110b	PO24	PO25	PO26	PO27	PO28	PO29	PO30	PO31
000111b	—	—	—	POE11#	—	POE4#	—	—
001010b	RXD4*1 SMISO4*1 SSCL4*1	TXD4*1 SMOSI4*1 SSDA4*1	CTS4#*1 RTS4#*1 SS4#*1	SCK4*1	—	SCK9	RXD9 SMISO9 SSCL9	TXD9 SMOSI9 SSDA9
001011b	RXD6 SMISO6 SSCL6	TXD6 SMOSI6 SSDA6	CTS6# RTS6# SS6#	SCK6	SS9# CTS9#	RTS9#	—	—
010001b	ET0_ERXD1	ET0_ERXD0	ET0_RX_CL K	ET0_RX_ER	ET0_TX_EN	ET0_ETXD0	ET0_ETXD1	ET0_CRS
010010b	RMII0_RXD1	RMII0_RXD0	REF50CK0	RMII0_RX_E R	RMII0_TXD_ EN	RMII0_TXD0	RMII0_TXD1	RMII0_CRS_ DV
100100b	—	—	—	—	SS11# CTS11# RTS11#	SCK11	SMISO11 SSCL11 RXD11	SMOSI11 SSDA11 TXD11
100101b	LCD_DATA0- B	LCD_TCON3 -B	LCD_TCON2 -B	LCD_TCON1 -B	LCD_TCON0 -B	LCD_CLK-B	—	—

—: Do not specify this value.

Note 1. This setting is not supported by 100-pin products.

23.2.14 PCn Pin Function Control Register (PCnPFS) (n = 0 to 7)

Address(es): PC0PFS 0008 C1A0h, PC1PFS 0008 C1A1h, PC2PFS 0008 C1A2h, PC3PFS 0008 C1A3h, PC4PFS 0008 C1A4h, PC5PFS 0008 C1A5h, PC6PFS 0008 C1A6h, PC7PFS 0008 C1A7h



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.19.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin PC0: IRQ14 (224/176/145/144/100 pin) PC1: IRQ12 (224/176/145/144/100 pin) PC6: IRQ13 (224/176/145/144/100 pin) PC7: IRQ14 (224/176/145/144/100 pin)	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Table 23.19 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-/100-Pin LFQFP, 145-Pin TFLGA

PSEL[5:0] Settings	Pin							
	PC0	PC1	PC2	PC3	PC4	PC5	PC6	PC7
000000b (initial value)	Hi-Z							
000001b	MTIOC3C	MTIOC3A	MTIOC4B	MTIOC4D	MTIOC3D	MTIOC3B	MTIOC3C	MTIOC3A
000010b	—	—	—	—	MTCLKC	MTCLKD	MTCLKA	MTCLKB
000011b	TCLKC	TCLKD	TCLKA	TCLKB	—	—	—	—
000101b	—	—	—	—	TMCI1	TMRI2	TMCI2	TMO2
000110b	PO17	PO18	PO21	PO24	PO25	PO29	PO30	PO31
000111b	—	—	—	—	POE0#	—	—	CACREF
001010b	—	SCK5	RXD5 SMISO5 SSCL5	TXD5 SMOSI5 SSDA5	SCK5	SCK8	RXD8 SMISO8 SSCL8	TXD8 SMOSI8 SSDA8
001011b	CTS5# RTS5# SS5#	—	—	—	SS8# CTS8#	RTS8#	—	—
001101b	SSLA1-A	SSLA2-A	SSLA3-A	—	SSLA0-A	RSPCKA-A	MOSIA-A	MISOA-A
010001b	ET0_ERXD3	ET0_ERXD2	ET0_RX_DV	ET0_TX_ER	ET0_TX_CLK	ET0_ETXD2	ET0_ETXD3	ET0_COL
011001b	—	—	MMC_CD-A*1	MMC_D0-A*1	MMC_D1-A*1	MMC_D5-A*1	MMC_D6-A*1	MMC_D7-A*1
011010b	—	—	SDHI_D3-A*1	SDHI_D0-A*1	SDHI_D1-A*1	—	—	—
011011b	—	—	—	QMO-A*1 QIO0-A*1	QMI-A*1 QIO1-A*1	—	—	—
011101b	—	—	—	—	—	—	TIC0	TOC0
011110b	—	—	GTIOC2B	GTIOC1B	GTETRGC	GTIOC1A	GTIOC3B	GTIOC3A
100100b	—	—	—	—	SS10# CTS10# RTS10#	SCK10	SMISO10 SSCL10 RXD10	SMOSI10 SSDA10 TXD10
100101b	—	LCD_DATA2 2-A*2	LCD_DATA1 9-A*2	LCD_DATA1 6-A*2	LCD_DATA1 5-A*2	LCD_DATA11 -A*2	LCD_DATA1 0-A*2	LCD_DATA9- A*2

—: Do not specify this value.

Note 1. This setting is not supported by 100-pin products.

Note 2. This setting is not supported by 145-/144-/100-pin products.

23.2.15 PDn Pin Function Control Register (PDnPFS) (n = 0 to 7)

Address(es): PD0PFS 0008 C1A8h, PD1PFS 0008 C1A9h, PD2PFS 0008 C1AAh, PD3PFS 0008 C1ABh,
PD4PFS 0008 C1ACh, PD5PFS 0008 C1ADh, PD6PFS 0008 C1AEh, PD7PFS 0008 C1AFh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.20.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin PD0: IRQ0 (224/176/145/144/100 pin) PD1: IRQ1 (224/176/145/144/100 pin) PD2: IRQ2 (224/176/145/144/100 pin) PD3: IRQ3 (224/176/145/144/100 pin) PD4: IRQ4 (224/176/145/144/100 pin) PD5: IRQ5 (224/176/145/144/100 pin) PD6: IRQ6 (224/176/145/144/100 pin) PD7: IRQ7 (224/176/145/144/100 pin)	R/W
b7	ASEL	Analog Input Function Select	0: Used other than as analog pin. 1: Used as analog pin. PD0: AN108 (224/176/145/144/100 pin) PD1: AN109 (224/176/145/144/100 pin) PD2: AN110 (224/176/145/144/100 pin) PD3: AN111 (224/176/145/144/100 pin) PD4: AN112 (224/176/145/144/100 pin) PD5: AN113 (224/176/145/144/100 pin) PD6: AN106 (224/176/145/144/100 pin) PD7: AN107 (224/176/145/144/100 pin)	R/W

Table 23.20 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-/100-Pin LFQFP, 145-Pin TFLGA

PSEL[5:0] Settings	Pin							
	PD0	PD1	PD2	PD3	PD4	PD5	PD6	PD7
000000b (initial value)	Hi-Z							
000001b	—	MTIOC4B	MTIOC4D	—	—	MTIC5W	MTIC5V	MTIC5U
000010b	—	—	—	—	—	MTCLKA	—	—
000111b	—	—	—	POE8#	POE11#	POE10#	POE4#	POE0#
001000b	POE4#	POE0#	—	MTIOC8D	MTIOC8B	MTIOC8C	MTIOC8A	—
001101b	—	MOSIC-A	MISOC-A	RSPCKC-A	SSLC0-A	SSLC1-A	SSLC2-A	SSLC3-A
010000b	—	CTX0	CRX0	—	—	—	—	—
010100b	—	—	ET1_EXOUT *2	ET1_WOL *2	ET1_MDIO*2	ET1_MDC *2	ET1_RX_CL K*1	ET1_RX_ER *1
010101b	—	—	—	—	—	—	REF50CK1*2	RMII1_RX_E R*2
011001b	—	—	MMC_D2-B	MMC_D3-B	MMC_CMD- B	MMC_CLK-B	MMC_D0-B	MMC_D1-B
011010b	—	—	SDHI_D2-B	SDHI_D3-B	SDHI_CMD- B	SDHI_CLK-B	SDHI_D0-B	SDHI_D1-B
011011b	—	—	QIO2-B	QIO3-B	QSSL-B	QSPCLK-B	QMO-B QIO0-B	QMI-B QIO1-B
011101b	—	—	TIC2	TOC2	—	—	—	—
011110b	GTIOC1B	GTIOC1A	GTIOC0B	GTIOC0A	—	—	—	—
100101b	LCD_EXTCL K-B	LCD_DATA2 3-B	LCD_DATA2 2-B	LCD_DATA2 1-B	LCD_DATA2 0-B	LCD_DATA1 9-B	LCD_DATA1 8-B	LCD_DATA1 7-B
101000b	—	—	—	—	PMGI1_MDI O*2	PMGI1_MDC *2	—	—

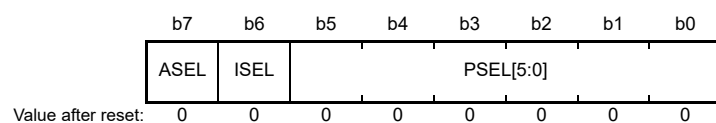
—: Do not specify this value.

Note 1. This setting is not supported by 145-/144-/100-pin products.

Note 2. This setting is not supported by 100-pin products.

23.2.16 PEn Pin Function Control Register (PEnPFS) (n = 0 to 7)

Address(es): PE0PFS 0008 C1B0h, PE1PFS 0008 C1B1h, PE2PFS 0008 C1B2h, PE3PFS 0008 C1B3h, PE4PFS 0008 C1B4h, PE5PFS 0008 C1B5h, PE6PFS 0008 C1B6h, PE7PFS 0008 C1B7h



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.21.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin PE2: IRQ7-DS (224/176/145/144/100 pin) PE5: IRQ5 (224/176/145/144/100 pin) PE6: IRQ6 (224/176/145/144/100 pin) PE7: IRQ7 (224/176/145/144/100 pin)	R/W
b7	ASEL	Analog Input Function Select	0: Used other than as analog pin. 1: Used as analog pin. PE0: ANEX0 (224/176/145/144/100 pin) PE1: ANEX1 (224/176/145/144/100 pin) PE2: AN100 (224/176/145/144/100 pin) PE3: AN101 (224/176/145/144/100 pin) PE4: AN102 (224/176/145/144/100 pin) PE5: AN103 (224/176/145/144/100 pin) PE6: AN104 (224/176/145/144/100 pin) PE7: AN105 (224/176/145/144/100 pin)	R/W

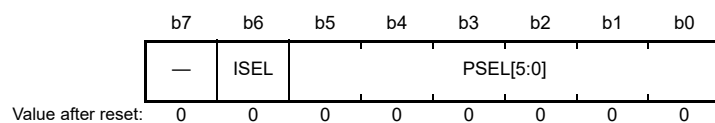
Table 23.21 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-/100-Pin LFQFP, 145-Pin TFLGA

PSEL[5:0] Settings	Pin							
	PE0	PE1	PE2	PE3	PE4	PE5	PE6	PE7
000000b (initial value)	Hi-Z							
000001b	—	MTIOC4C	MTIOC4A	MTIOC4B	MTIOC4D	MTIOC4C	—	—
000010b	—	—	—	—	MTIOC1A	MTIOC2B	—	—
000110b	—	PO18	PO23	PO26	PO28	—	—	—
000111b	—	—	—	POE8#	—	—	—	—
001000b	MTIOC3D	MTIOC3B	—	—	—	—	MTIOC6C	MTIOC6A
001100b	SCK12	TXD12 SMOSI12 SSDA12 TXDX12 SIOX12	RXD12 SMISO12 SSCL12 RXDX12	CTS12# RTS12# SS12#	—	—	—	—
001101b	SSLB1-B	SSLB2-B	SSLB3-B	—	SSLB0-B	RSPCKB-B	MOSIB-B	MISOB-B
010001b	—	—	—	ET0_ERXD3	ET0_ERXD2	ET0_RX_CLK	—	—
010010b	—	—	—	—	—	REF50CK0	—	—
011001b	MMC_D4-B	MMC_D5-B	MMC_D6-B	MMC_D7-B	—	—	MMC_CD-B	MMC_RES#-B
011010b	—	—	—	—	—	—	SDHI_CD	SDHI_WP
011011b	—	—	TIC3	TOC3	—	—	TIC1	TOC1
011110b	GTIOC2B	GTIOC1B	GTIOC0B	GTIOC2A	GTIOC1A	GTIOC0A	GTIOC3B	GTIOC3A
100101b	LCD_DATA16-B	LCD_DATA15-B	LCD_DATA14-B	LCD_DATA13-B	LCD_DATA12-B	LCD_DATA11-B	LCD_DATA10-B	LCD_DATA9-B

—: Do not specify this value.

23.2.17 PF_n Pin Function Control Register (PF_nPFS) (n = 0 to 2, 5)

Address(es): PF0PFS 0008 C1B8h, PF1PFS 0008 C1B9h, PF2PFS 0008 C1BAh, PF5PFS 0008 C1BDh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.22 and Table 23.23.	R/W
b6	ISEL	Interrupt Input Function Select	0: Not used as IRQn input pin 1: Used as IRQn input pin PF5: IRQ4 (224/176/145/144 pin)	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Table 23.22 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-Pin LQFP

PSEL[5:0] Settings	Pin			
	PF0	PF1	PF2	PF5
000000b (initial value)	Hi-Z			
001010b	TXD1 SMOSI1 SSDA1	SCK1	RXD1 SMISO1 SSCL1	—
010111b	—	—	—	SSILRCK0

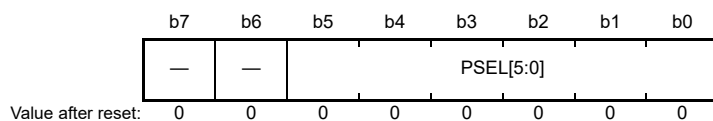
—: Do not specify this value.

Table 23.23 Register Settings for Input/Output Pin Function in 145-Pin TFLGA, 144-Pin LQFP

PSEL[5:0] Settings	Pin
	PF5
000000b (initial value)	Hi-Z
010111b	SSILRCK0

23.2.18 PGn Pin Function Control Register (PGnPFS) (n = 0 to 7)

Address(es): PG0PFS 0008 C1C0h, PG1PFS 0008 C1C1h, PG2PFS 0008 C1C2h, PG3PFS 0008 C1C3h, PG4PFS 0008 C1C4h, PG5PFS 0008 C1C5h, PG6PFS 0008 C1C6h, PG7PFS 0008 C1C7h



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.24.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

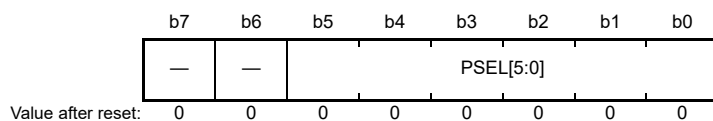
Table 23.24 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-Pin LFQFP

PSEL[5:0] Settings	Pin							
	PG0	PG1	PG2	PG3	PG4	PG5	PG6	PG7
000000b (initial value)	Hi-Z							
010100b	ET1_RX_CLK	ET1_RX_ER	ET1_TX_CLK	ET1_ETXD0	ET1_ETXD1	ET1_ETXD2	ET1_ETXD3	ET1_TX_ER
010101b	REF50CK1	RMII1_RX_ER	—	RMII1_TXD0	RMII1_TXD1	—	—	—

—: Do not specify this value.

23.2.19 PHn Pin Function Control Register (PHnPFS) (n = 0 to 7)

Address(es): PH0PFS 0008 C1C8h, PH1PFS 0008 C1C9h, PH2PFS 0008 C1CAh, PH3PFS 0008 C1CBh,
PH4PFS 0008 C1CCh, PH5PFS 0008 C1CDh, PH6PFS 0008 C1CEh, PH7PFS 0008 C1CFh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.25.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

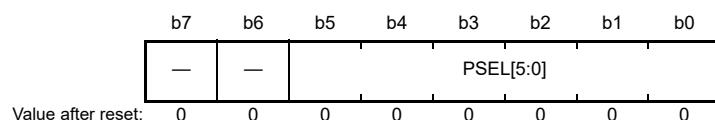
Table 23.25 Register Settings for Input/Output Pin Function in 224-Pin LFBGA

PSEL[5:0] Settings	Pin							
	PH0	PH1	PH2	PH3	PH4	PH5	PH6	PH7
000000b (initial value)	Hi-Z							
001010b	SCK7	SMISO7 SSCL7 RXD7	SMOSI7 SSDA7 TXD7	—	—	—	—	—
001011b	—	—	—	RTS7# CTS7# SS7#	—	—	—	—
001101b	RSPCKA	MOSIA	MISOA	SSLA0	SSLA1	SSLA2	SSLA3	—
011101b	TIC0	TOC0	—	—	—	—	—	—
011110b	GTETRGA	GTETRGB	GTETRGC	GTETRGD	GTADSM0	GTADSM1	GTIOC0A	GTIOC0B
101010b	—	—	—	—	—	—	CLKOUT	CLKOUT25M

—: Do not specify this value.

23.2.20 PJn Pin Function Control Register (PJnPFS) (n = 0 to 3, 5)

Address(es): PJ0PFS 0008 C1D0h, PJ1PFS 0008 C1D1h, PJ2PFS 0008 C1D2h, PJ3PFS 0008 C1D3h, PJ5PFS 0008 C1D5h



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.26 to Table 23.28.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Table 23.26 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-Pin LFQFP

PSEL[5:0] Settings	Pin				
	PJ0	PJ1	PJ2	PJ3	PJ5
000000b (initial value)	Hi-Z				
000001b	MTIOC6B	MTIOC6A	—	MTIOC3C	—
001010b	SCK8	RXD8 SMISO8 SSCL8	TXD8 SMOSI8 SSDA8	CTS6# RTS6# SS6#	—
001011b	—	—	—	CTS0# RTS0# SS0#	CTS2# RTS2# SS2#
001101b	SSLC1-B	SSLC2-B	SSLC3-B	—	—
010001b	EPLSOUT0	EPLSOUT1	—	ET0_EXOUT	EPLSOUT0
010111b	—	—	—	SSITXD0	SSIRXD0
011000b	—	—	—	EDACK1	—
011011b	—	—	—	QMO-C*1 QIO0-C*1	QMI-C*1 QIO1-C*1
100001b	—	—	—	—	POE8#
100101b	LCD_DATA0-A	LCD_TCON3-A	LCD_TCON2-A	—	—
101010b	—	—	CLKOUT25M	—	—

—: Do not specify this value.

Note 1. This setting is not supported by 176-pin products.

Table 23.27 Register Settings for Input/Output Pin Function in 145-Pin TFLGA, 144-Pin LFQFP

PSEL[5:0] Settings	Pin	
	PJ3	PJ5
000000b (initial value)	Hi-Z	
000001b	MTIOC3C	—
001010b	CTS6# RTS6# SS6#	—
001011b	CTS0# RTS0# SS0#	CTS2# RTS2# SS2#
010001b	ET0_EXOUT	EPLSOUT0
010111b	SSITXD0	SSIRXD0
011000b	EDACK1	—
100001b	—	POE8#

—: Do not specify this value.

Table 23.28 Register Settings for Input/Output Pin Function in 100-Pin LFQFP

PSEL[5:0] Settings	Pin
	PJ3
000000b (initial value)	Hi-Z
000001b	MTIOC3C
001010b	CTS6# RTS6# SS6#
001011b	CTS0# RTS0# SS0#
010001b	ET0_EXOUT
010111b	SSITXD0
011000b	EDACK1

23.2.21 P_Kn Pin Function Control Register (P_KnPFS) (n = 0 to 7)

Address(es): PK0PFS 0008 C1D8h, PK1PFS 0008 C1D9h, PK2PFS 0008 C1DAh, PK3PFS 0008 C1DBh, PK4PFS 0008 C1DCh, PK5PFS 0008 C1DDh, PK6PFS 0008 C1DEh, PK7PFS 0008 C1DFh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.29.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Table 23.29 Register Settings for Input/Output Pin Function in 224-Pin LFBGA

PSEL[5:0] Settings	Pin							
	PK0	PK1	PK2	PK3	PK4	PK5	PK6	PK7
000000b (initial value)	Hi-Z							
001010b	SCK8	SMISO8 SSCL8 RXD8	SMOSI8 SSDA8 TXD8	—	—	—	—	—
001011b	—	—	—	RTS8# CTS8# SS8#	—	—	—	—
001101b	RSPCKB	MOSIB	MISOB	SSLB0	SSLB1	SSLB2	SSLB3	—
010001b	ET0_MDC	ET0_COL	ET0_RX_DV	ET0_TX_ER	ET0_ERXD2	ET0_ERXD3	—	—
011101b	TIC1	TOC1	—	—	—	—	—	—
011110b	GTETRGA	GTETRGB	GTETRGC	GTETRGD	GTADSM0	GTADSM1	GTIOC1A	GTIOC1B
101000b	PMGIO_MDC	—	—	—	—	—	—	—

—: Do not specify this value.

23.2.22 PLn Pin Function Control Register (PLnPFS) (n = 0 to 7)

Address(es): PLOPFS 0008 C1E0h, PL1PFS 0008 C1E1h, PL2PFS 0008 C1E2h, PL3PFS 0008 C1E3h, PL4PFS 0008 C1E4h, PL5PFS 0008 C1E5h, PL6PFS 0008 C1E6h, PL7PFS 0008 C1E7h



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.30.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Table 23.30 Register Settings for Input/Output Pin Function in 224-Pin LFBGA

PSEL[5:0] Settings	Pin							
	PL0	PL1	PL2	PL3	PL4	PL5	PL6	PL7
000000b (initial value)	Hi-Z							
001010b	SCK9	SMISO9 SSCL9 RXD9	SMOSI9 SSDA9 TXD9	—	—	—	—	—
001011b	—	—	—	RTS9# CTS9# SS9#	—	—	—	—
001101b	RSPCKC	MOSIC	MISOC	SSLC0	SSLC1	SSLC2	SSLC3	—
010001b	ET0_ERXD0	ET0_ERXD1	ET0_RX_ER	ET0_RX_CLK	ET0_ETXD0	ET0_ETXD1	ET0_TX_EN	ET0_MDIO
010010b	RMII0_RXD0	RMII0_RXD1	RMII0_RX_ER	REF50CK0	RMII0_TXD0	RMII0_TXD1	RMII0_TXD_EN	—
011101b	TIC2	TOC2	—	—	—	—	—	—
011110b	GTETRGA	GTETRGB	GTETRGC	GTETRGD	GTADSM0	GTADSM1	GTIOC2A	GTIOC2B
101000b	—	—	—	—	—	—	—	PMGIO_MDI O

—: Do not specify this value.

23.2.23 PMn Pin Function Control Register (PMnPFS) (n = 0 to 7)

Address(es): PM0PFS 0008 C1E8h, PM1PFS 0008 C1E9h, PM2PFS 0008 C1EAh, PM3PFS 0008 C1EBh,
PM4PFS 0008 C1ECh, PM5PFS 0008 C1EDh, PM6PFS 0008 C1EEh, PM7PFS 0008 C1EFh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.31.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

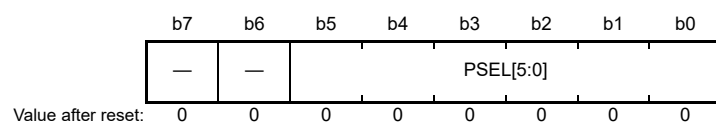
Table 23.31 Register Settings for Input/Output Pin Function in 224-Pin LFBGA

PSEL[5:0] Settings	Pin							
	PM0	PM1	PM2	PM3	PM4	PM5	PM6	PM7
000000b (initial value)	Hi-Z							
001010b	SCK10	SMISO10 SSCL10 RXD10	SMOS10 SSDA10 TXD10	—	—	—	—	—
001011b	—	—	—	RTS10# CTS10# SS10#	—	—	—	—
010001b	—	—	—	—	ET0_ETXD2	ET0_ETXD3	ET0_TX_CLK	ET0_CRS
010010b	—	—	—	—	—	—	—	RMII0_CRS_DV
010100b	ET1_ERXD0	ET1_ERXD1	ET1_ERXD2	ET1_ERXD3	—	—	—	—
010101b	RMII1_RXD0	RMII1_RXD1	—	—	—	—	—	—
011010b	SDHI_CLK-D	SDHI_CMD-D	SDHI_D0-D	SDHI_D1-D	SDHI_D2-D	SDHI_D3-D	SDHI_CD	SDHI_WP
011011b	QSPCLK-A	QSSL-A	QMO-A QIO0-A	QMI-A QIO1-A	QIO2-A	QIO3-A	—	—
011101b	TIC3	TOC3	—	—	—	—	—	—
011110b	GTETRGA	GTETRGB	GTETRG	GTETRGC	GTETRGD	GTADSM0	GTADSM1	GTIOC3A GTIOC3B

—: Do not specify this value.

23.2.24 P_N Pin Function Control Register (P_NPFS) (n = 0 to 5)

Address(es): PN0PFS 0008 C1F0h, PN1PFS 0008 C1F1h, PN2PFS 0008 C1F2h, PN3PFS 0008 C1F3h, PN4PFS 0008 C1F4h, PN5PFS 0008 C1F5h



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.32.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Table 23.32 Register Settings for Input/Output Pin Function in 224-Pin LFBGA

PSEL[5:0] Settings	Pin					
	PN0	PN1	PN2	PN3	PN4	PN5
000000b (initial value)	Hi-Z					
010100b	ET1_ETXD2	ET1_ETXD3	ET1_TX_CLK	ET1_RX_ER	ET1_MDIO	ET1_MDC
010101b	—	—	—	RMII1_RX_ER	—	—
011011b	—	—	—	—	QSPCLK-C	QSSL-C
101000b	—	—	—	—	PMGI1_MDI_O	PMGI1_MDC

—: Do not specify this value.

23.2.25 PQn Pin Function Control Register (PQnPFS) (n = 0 to 7)

Address(es): PQ0PFS 0008 C1F8h, PQ1PFS 0008 C1F9h, PQ2PFS 0008 C1FAh, PQ3PFS 0008 C1FBh, PQ4PFS 0008 C1FCh, PQ5PFS 0008 C1FDh, PQ6PFS 0008 C1FEh, PQ7PFS 0008 C1FFh



Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PSEL[5:0]	Pin Function Select	These bits select the peripheral function. For individual pin functions, refer to Table 23.33.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Table 23.33 Register Settings for Input/Output Pin Function in 224-Pin LFBGA

PSEL[5:0] Settings	Pin							
	PQ0	PQ1	PQ2	PQ3	PQ4	PQ5	PQ6	PQ7
000000b (initial value)	Hi-Z							
001010b	SCK11	SMISO11 SSCL11 RXD11	SMOSI11 SSDA11 TXD11	—	—	—	—	—
001011b	—	—	—	RTS11# CTS11# SS11#	—	—	—	—
010100b	ET1_CRS	ET1_COL	ET1_RX_DV	ET1_TX_ER	ET1_RX_CLK	ET1_ETXD0	ET1_ETXD1	ET1_TX_EN
010101b	RMII1_CRS_DV	—	—	—	REF50CK1	RMII1_TXD0	RMII1_TXD1	RMII1_TXD_EN

—: Do not specify this value.

23.2.26 CS Output Enable Register (PFCSE)

Address(es): 0008 C100h

b7	b6	b5	b4	b3	b2	b1	b0
CS7E	CS6E	CS5E	CS4E	CS3E	CS2E	CS1E	CS0E
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	CS0E	CS0 Enable	0: Disables CSn# output. 1: Enables CSn# output.	R/W
b1	CS1E	CS1 Enable	(n = 0 to 7)	R/W
b2	CS2E	CS2 Enable		R/W
b3	CS3E	CS3 Enable		R/W
b4	CS4E	CS4 Enable		R/W
b5	CS5E	CS5 Enable		R/W
b6	CS6E	CS6 Enable		R/W
b7	CS7E	CS7 Enable		R/W

CSnE Bit (CSn Enable) (n = 0 to 7)

These bits enable or disable the corresponding pins to output the CSn# signal.

To enable output of the CSn# signal, set the corresponding CSnE bit in PFCSE to 1.

23.2.27 CS Output Pin Select Register 0 (PFCSS0)

Address(es): 0008 C102h

b7	b6	b5	b4	b3	b2	b1	b0
CS3S[1:0]	CS2S[1:0]	CS1S[1:0]	—	CS0S			
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	CS0S	CS0# Output Pin Select*1	0: Set P60 as CS0# output pin 1: Set PC7 as CS0# output pin	R/W
b1	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b3, b2	CS1S[1:0]	CS1# Output Pin Select*2	b3 b2 0 0: Set P61 as CS1# output pin 0 1: Set P71 as CS1# output pin 1 x: Set PC6 as CS1# output pin	R/W
b5, b4	CS2S[1:0]	CS2# Output Pin Select*3	b5 b4 0 0: Set P62 as CS2# output pin 0 1: Set P72 as CS2# output pin 1 x: Set PC5 as CS2# output pin	R/W
b7, b6	CS3S[1:0]	CS3# Output Pin Select*4	b7 b6 0 0: Set P63 as CS3# output pin 0 1: Set P73 as CS3# output pin 1 x: Set PC4 as CS3# output pin	R/W

x: Don't care

Note 1. P60 is not present in 100-pin products. When CS0# output is used, set this bit to 1.

Note 2. P61 and P71 are not present in 100-pin products. When CS1# output is used, set these bits to 1xb.

Note 3. P62 and P72 are not present in 100-pin products. When CS2# output is used, set these bits to 1xb.

Note 4. P63 and P73 are not present in 100-pin products. When CS3# output is used, set these bits to 1xb.

CS0S Bit (CS0# Output Pin Select)**CSnS[1:0] Bits (CSn# Output Pin Select) (n = 1 to 3)**

When CSn# output is enabled (the PFCSE.CSnE bit = 1), the CSn# output pin is selected.

23.2.28 CS Output Pin Select Register 1 (PFCSS1)

Address(es): 0008 C103h

b7	b6	b5	b4	b3	b2	b1	b0
CS7S[1:0]	CS6S[1:0]	CS5S[1:0]	CS4S[1:0]				
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	CS4S[1:0]	CS4# Output Pin Select*1	b1 b0 0 0: Set P64 as CS4# output pin 0 1: Set P74 as CS4# output pin 1 x: Set P24 as CS4# output pin	R/W
b3, b2	CS5S[1:0]	CS5# Output Pin Select*2	b3 b2 0 0: Set P65 as CS5# output pin 0 1: Set P75 as CS5# output pin 1 x: Set P25 as CS5# output pin	R/W
b5, b4	CS6S[1:0]	CS6# Output Pin Select*3	b5 b4 0 0: Set P66 as CS6# output pin 0 1: Set P76 as CS6# output pin 1 x: Set P26 as CS6# output pin	R/W
b7, b6	CS7S[1:0]	CS7# Output Pin Select*4	b7 b6 0 0: Set P67 as CS7# output pin 0 1: Set P77 as CS7# output pin 1 x: Set P27 as CS7# output pin	R/W

x: Don't care

Note 1. P64 and P74 are not present in 100-pin products. When CS4# output is used, set these bits to 1xb.

Note 2. P65 and P75 are not present in 100-pin products. When CS5# output is used, set these bits to 1xb.

Note 3. P66 and P76 are not present in 100-pin products. When CS6# output is used, set these bits to 1xb.

Note 4. P67 and P77 are not present in 100-pin products. When CS7# output is used, set these bits to 1xb.

CSnS[1:0] Bits (CSn# Output Pin Select) (n = 4 to 7)

When CSn# output is enabled (the PFCSE.CSNE bit = 1), the CSn# output pin is selected.

23.2.29 Address Output Enable Register 0 (PFAOE0)

Address(es): 0008 C104h

	b7	b6	b5	b4	b3	b2	b1	b0
	A15E	A14E	A13E	A12E	A11E	A10E	A9E	A8E
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	A8E	Address A8 Output Enable	0: Disables A8 output. 1: Enables A8 output.	R/W
b1	A9E	Address A9 Output Enable	0: Disables A9 output. 1: Enables A9 output.	R/W
b2	A10E	Address A10 Output Enable	0: Disables A10 output. 1: Enables A10 output.	R/W
b3	A11E	Address A11 Output Enable	0: Disables A11 output. 1: Enables A11 output.	R/W
b4	A12E	Address A12 Output Enable	0: Disables A12 output. 1: Enables A12 output.	R/W
b5	A13E	Address A13 Output Enable	0: Disables A13 output. 1: Enables A13 output.	R/W
b6	A14E	Address A14 Output Enable	0: Disables A14 output. 1: Enables A14 output.	R/W
b7	A15E	Address A15 Output Enable	0: Disables A15 output. 1: Enables A15 output.	R/W

23.2.30 Address Output Enable Register 1 (PFAOE1)

Address(es): 0008 C105h

	b7	b6	b5	b4	b3	b2	b1	b0
	A23E	A22E	A21E	A20E	A19E	A18E	A17E	A16E
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	A16E	Address A16 Output Enable	0: Disables A16 output. 1: Enables A16 output.	R/W
b1	A17E	Address A17 Output Enable	0: Disables A17 output. 1: Enables A17 output.	R/W
b2	A18E	Address A18 Output Enable	0: Disables A18 output. 1: Enables A18 output.	R/W
b3	A19E	Address A19 Output Enable	0: Disables A19 output. 1: Enables A19 output.	R/W
b4	A20E	Address A20 Output Enable	0: Disables A20 output. 1: Enables A20 output.	R/W
b5	A21E	Address A21 Output Enable	0: Disables A21 output. 1: Enables A21 output.	R/W
b6	A22E	Address A22 Output Enable	0: Disables A22 output. 1: Enables A22 output.	R/W
b7	A23E	Address A23 Output Enable	0: Disables A23 output. 1: Enables A23 output.	R/W

23.2.31 External Bus Control Register 0 (PFBCR0)

Address(es): 0008 C106h

b7	b6	b5	b4	b3	b2	b1	b0
WR32B C32E	WR1B C1E	DH32E	DHE	BCLKO	ADRH MS2	ADRH MS	ADRLE
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ADRLE	A0 to A7 Output Enable	0: Configures PA0 to PA7 as the I/O port pins. 1: Configures PA0 to PA7 as the external address bus A0 to A7.	R/W
b1	ADRHMS	A16 to A23 Output Enable	Refer to Table 23.34.	R/W
b2	ADRHMS2	A16 to A23 Output Enable 2		R/W
b3	BCLKO	BCLK Forced Output	0: BCLK is output when EXBE = 1 and not output when EXBE = 0. 1: BCLK is output regardless of the setting of EXBE.	R/W
b4	DHE	D8 to D15 Output Enable	0: Configures PE0 to PE7 as the I/O port pins. 1: Configures PE0 to PE7 as the external data bus D8 to D15.	R/W
b5	DH32E	D16 to D31 Output Enable	0: Set P90 to P97, and PG0 to PG7 as I/O port 1: Set P90 to P97, and PG0 to PG7 as external data bus D16 to D31.	R/W
b6	WR1BC1E	WR1#/BC1# Output Enable	0: Configures P51 as the I/O port pin. 1: Configures P51 as the WR1# or BC1# pin.	R/W
b7	WR32BC32E	WR3#/BC3# and WR2#/ BC2# Output Enable	0: Set P13 and P12 as I/O port 1: Set P13 as WR#2 or BC2#, and set P12 as WR#3 or BC3#	R/W

BCLKO Bit (BCLK Forced Output)

This bit enables or disables forced output on the BCLK pin.

When this bit is set to 0, BCLK output is enabled or disabled according to the setting of the EXBE bit; when this bit is 1, BCLK is output regardless of the setting of the EXBE bit.

Note that if the bit is set to 1, BCLK is output regardless of the PMR register.

[Setting procedure]

Output enabled: PSTOP1 (stopped) → BCLKO = 1 → PSTOP1 (operating)

Output disabled: PSTOP1 (operating) → PSTOP1 (stopped) → BCLKO = 0

Table 23.34 Settings for External Address Buses A16 to A23

ADRHMS Bit	ADRHMS2 Bit	Settings for External Address Buses A16 to A23
0	0	Set PC0 to PC7.
0	1	Set PC0, PC1, P71, P72, P74, and PC5 to PC7.
1	0	In 224-pin and 176-pin products: set P90 to P97. In 145-pin and 144-pin products: set P90 to P93. (Address bits A20 to A23 are not allocated.)
1	1	Setting prohibited

Note: The setting must be 00b for 100-pin products.

23.2.32 External Bus Control Register 1 (PFBCR1)

Address(es): 0008 C107h

b7	b6	b5	b4	b3	b2	b1	b0
SDCLK E	DQM1E	—	MDSD E	ALES	ALEOE	WAITS[1:0]	
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	WAITS[1:0]	WAIT Select	Refer to Table 23.35.	R/W
b2	ALEOE	ALE Output Enable	0: Disables ALE pin output. 1: Enables ALE pin output.	R/W
b3	ALES	ALE Select	0: Configures P54 as the ALE pin.*1 1: Configures P10 as the ALE pin.	R/W
b4	MDSDE	SDRAM Pin Enable*2	0: Disables SDRAM pin output. (CKE, SDCS#, RAS#, CAS#, WE#, and DQM0) 1: Enables SDRAM pin output. (CKE, SDCS#, RAS#, CAS#, WE#, and DQM0)	R/W
b5	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b6	DQM1E	DQM1 Enable*2	0: Disables DQM1 output 1: Enables DQM1 output	R/W
b7	SDCLKE	SDCLK Enable*2	0: Disables SDCLK output 1: Enables SDCLK output	R/W

Note 1. Set this bit to 0 for the products other than 224- and 176-pin products.

Note 2. This bit is reserved for 100-pin products. It is read as 0. The write value should be 0.

WAITS[1:0] Bits (WAIT Select)

The port pin specified by the WAITS[1:0] bits and PFBCR3.WAITS2 bit become the WAIT# pin when the external bus is enabled. However, if the specified port pin is not to be used as the WAIT# pin, the external wait enable bit (EWENB) in the CSn mode register (CSnMOD) can be cleared (disabling external wait) to make the pin available for use as a general input port pin.

If the specified WAIT# pin is not to be used as the WAIT input or as a general input port pin, pull the level on the WAIT# pin up or down.

ALEOE Bit (ALE Output Enable)

This bit enables or disables output of the ALE pin.

ALES Bit (ALE Select)

This bit selects the pins to output ALE.

To modify this bit, make sure that the ALEOE bit is set to 0. Furthermore, if the ALE pin is to be used, set this bit to 0 for the products other than 224- and 176-pin products.

MDSDE Bit (SDRAM Pin Enable)

This bit enables or disables output of the SDRAM pin (CKE, SDCS#, RAS#, CAS#, WE#, and DQM0).

The DQM1 pin is enabled or disabled individually by the DQM1E bit, while the MDSDE bit is 1. The SDCLK pin is enabled or disabled independently by the SDCLKE bit, regardless of the MDSDE setting.

DQM1E Bit (DQM1 Enable)

This bit enables or disables output of the DQM1 pin.

When the MDSDE bit is set to 1, setting of the DQM1E bit is enabled. When the MDSDE bit is set to 0, setting of the

DQM1E bit has no effect.

SDCLKE Bit (SDCLK Enable)

This bit enables or disables output of the SDCLK pin.

The SCKCR.PSTOP0 bit should be set to 1 before changing the setting of the SDCLKE bit.

Table 23.35 Setting of the WAIT Pin

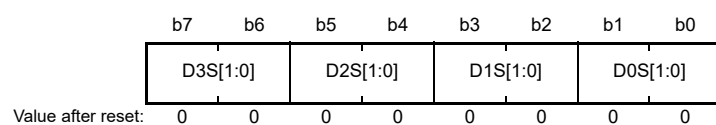
PFBCR3.WAITS2 Bit	PFBCR1.WAITS[1:0] Bits	Setting of the WAIT Pin
0	00	Setting prohibited
0	01	P55 is set as the WAIT# input pin.
0	10	PC5 is set as the WAIT# input pin.
0	11	P51 is set as the WAIT# input pin.
1	00	PF5 is set as the WAIT# input pin.*1
1	01	Setting prohibited
1	1x	Setting prohibited

x: Don't care

Note 1. This setting is not supported by 100-pin products (setting prohibited).

23.2.33 External Bus Control Register 2 (PFBCR2)

Address(es): 0008 C108h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	D0S[1:0]	D0 Selection	b1 b0 0 0: PD0 is set as D0 pin. 0 1: PE0 is set as D0 pin.*1 1 0: P55 is set as D0 pin. 1 1: P61 is set as D0 pin.	R/W
b3, b2	D1S[1:0]	D1 Selection	b3 b2 0 0: PD1 is set as D1 pin. 0 1: PE1 is set as D1 pin.*1 1 0: P54 is set as D1 pin. 1 1: P62 is set as D1 pin.	R/W
b5, b4	D2S[1:0]	D2 Selection	b5 b4 0 0: PD2 is set as D2 pin. 0 1: PE2 is set as D2 pin.*1 1 0: PC6 is set as D2 pin. 1 1: P63 is set as D2 pin.	R/W
b7, b6	D3S[1:0]	D3 Selection	b7 b6 0 0: PD3 is set as D3 pin. 0 1: PE3 is set as D3 pin.*1 1 0: PC5 is set as D3 pin. 1 1: P64 is set as D3 pin.	R/W

Note 1. Do not make this setting if PE0 to PE7 are to be set to operate as external data bus pins D8 to D15 and the setting of PFBCR0.DHE is 1.

23.2.34 External Bus Control Register 3 (PFBCR3)

Address(es): 0008 C109h

b7	b6	b5	b4	b3	b2	b1	b0
WAITS 2	SDCLK DRV	—	—	—	—	—	DLHS

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	DLHS	D4 to D7 Selection	0: PD4 to PD7 are set as pins D4 to D7 1: PE4 to PE7 are set as pins D4 to D7.	R/W
b5 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b6	SDCLKDRV	SDCLK Pin Driving Ability Select	This bit controls the driving capacity of the SDCLK pin. The selection is based on the combination of this setting and that of the PFBCR1.SDCLKE bit. SDCLKE SDCLKDRV 0 0 : SDCLK pin output is disabled. 0 1 : Do not set this value. 1 0 : Use the pin with the SDCLK set for a frequency no higher than 60 MHz. 1 1 : Use the pin with SDCLK set for a frequency above 60 MHz.	R/W
b7	WAITS2	WAIT Select 2	Refer to Table 23.35.	R/W

23.2.35 Ethernet Control Register (PFENET)

Address(es): 0008 C10Eh

b7	b6	b5	b4	b3	b2	b1	b0
—	—	PHYMODE1	PHYMODE0	—	—	—	—
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	PHYMODE0	Ethernet Channel 0 Mode Set	0: RMII mode (ETHERC0) 1: MII mode (ETHERC0)	R/W
b5	PHYMODE1	Ethernet Channel 1 Mode Set	0: RMII mode (ETHERC1)*1 1: MII mode (ETHERC1)	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Set this bit to 0 in 145-/144-/100-pin products.

PHYMODE0 Bit (Ethernet Channel 0 Mode Set)

This bit specifies the PHY mode of the ETHERC0.

Select the same mode as the one specified by the pin function select bits (PmnPFS.PSEL[5:0]).

When the signals for the RMII mode have been specified by the PmnPFS.PSEL[5:0] bits, set the PHYMODE bit to 0 (RMII mode), whereas when the signals for the MII mode have been specified by the PmnPFS.PSEL[5:0] bits, set the PHYMODE bit to 1 (MII mode).

PHYMODE1 Bit (Ethernet Channel 1 Mode Set)

This bit specifies the PHY mode of the ETHERC1.

Select the same mode as the one specified by the pin function select bits (PmnPFS.PSEL[5:0]).

When the signals for the RMII mode have been specified by the PmnPFS.PSEL[5:0] bits, set the PHYMODE bit to 0 (RMII mode), whereas when the signals for the MII mode have been specified by the PmnPFS.PSEL[5:0] bits, set the PHYMODE bit to 1 (MII mode).

23.3 How to Set the External Bus Interface

If the external bus interface is to be used, set the MPC registers according to Table 23.36 and then set the external bus enable bit (EXBE) in system control register 0 (SYSCR0) to 1.

Table 23.36 lists how to set up port pins to act as the external bus interface. For details on the relevant registers of the MPC, refer to section 23.2, Register Descriptions.

Table 23.36 How to Set the External Bus Interface (1/4)

Port	Output Signal	Settings of MPC Registers		
		224-Pin, 176-Pin	145-Pin, 144-Pin	100-Pin
P10	ALE	PFBCR1.ALEOE = 1, PFBCR1.ALES = 1	(not provided)	(not provided)
P12	WR3#/BC3#	PFBCR0.WR32BC32E = 1	(not supported)	(not supported)
P13	WR2#/BC2#	PFBCR0.WR32BC32E = 1	(not supported)	(not supported)
P24	CS4#	PFCSE.CS4E = 1, PFCSS1.CS4S[1:0] = 10/11		
P25	CS5#	PFCSE.CS5E = 1, PFCSS1.CS5S[1:0] = 10/11		
P26	CS6#	PFCSE.CS6E = 1, PFCSS1.CS6S[1:0] = 10/11		
P27	CS7#	PFCSE.CS7E = 1, PFCSS1.CS7S[1:0] = 10/11		
P50	WR0#/WR#	—		
P51	WR1#/BC1#	PFBCR0.WR1BC1E = 1		
	WAIT#	PFBCR1.WAITS[1:0] = 11, PFBCR3.WAITS2 = 0		
P52	RD#	—		
P53	BCLK	—		
P54	ALE	PFBCR1.ALEOE = 1, PFBCR1.ALES = 0		
	D1[A1/D1]	PFBCR2.D1S[1:0] = 10		
P55	WAIT#	PFBCR1.WAITS[1:0] = 01, PFBCR3.WAITS2 = 0		
	D0[A0/D0]	PFBCR2.D0S[1:0] = 10		
P60	CS0#	PFCSE.CS0E = 1, PFCSS0.CS0S = 0		(not provided)
P61	CS1#	PFCSE.CS1E = 1, PFCSS0.CS1S[1:0] = 00		(not provided)
	SDCS#	PFBCR1.MDSDE = 1		(not provided)
	D0[A0/D0]	PFBCR2.D0S[1:0] = 11		(not provided)
P62	CS2#	PFCSE.CS2E = 1, PFCSS0.CS2S[1:0] = 00		(not provided)
	RAS#	PFBCR1.MDSDE = 1		(not provided)
	D1[A1/D1]	PFBCR2.D1S[1:0] = 11		(not provided)
P63	CS3#	PFCSE.CS3E = 1, PFCSS0.CS3S[1:0] = 00		(not provided)
	CAS#	PFBCR1.MDSDE = 1		(not provided)
	D2[A2/D2]	PFBCR2.D2S[1:0] = 11		(not provided)
P64	CS4#	PFCSE.CS4E = 1, PFCSS1.CS4S[1:0] = 00		(not provided)
	WE#	PFBCR1.MDSDE = 1		(not provided)
	D3[A3/D3]	PFBCR2.D3S[1:0] = 11		(not provided)
P65	CS5#	PFCSE.CS5E = 1, PFCSS1.CS5S[1:0] = 00		(not provided)
	CKE	PFBCR1.MDSDE = 1		(not provided)
P66	CS6#	PFCSE.CS6E = 1, PFCSS1.CS6S[1:0] = 00		(not provided)
	DQM0	PFBCR1.MDSDE = 1		(not provided)
P67	CS7#	PFCSE.CS7E = 1, PFCSS1.CS7S[1:0] = 00		(not provided)
	DQM1	PFBCR1.MDSDE = 1, PFBCR1.DQM1E = 1		(not provided)
P70	SDCLK	PFBCR1.SDCLKE = 1		(not provided)
P71	CS1#	PFCSE.CS1E = 1, PFCSS0.CS1S[1:0] = 01		(not provided)
	A18	PFAOE1.A18E = 1, PFBCR0.ADRHMS = 0, PFBCR0.ADRHMS2 = 1		(not provided)

Table 23.36 How to Set the External Bus Interface (2/4)

Port	Output Signal	Settings of MPC Registers		
		224-Pin, 176-Pin	145-Pin, 144-Pin	100-Pin
P72	CS2#	PFCSE.CS2E = 1, PFCSS0.CS2S[1:0] = 01		(not provided)
	A19	PFAOE1.A19E = 1, PFBCR0.ADRHMS = 0, PFBCR0.ADRHMS2 = 1		(not provided)
P73	CS3#	PFCSE.CS3E = 1, PFCSS0.CS3S[1:0] = 01		(not provided)
P74	CS4#	PFCSE.CS4E = 1, PFCSS1.CS4S[1:0] = 01		(not provided)
	A20	PFAOE1.A20E = 1, PFBCR0.ADRHMS = 0, PFBCR0.ADRHMS2 = 1		(not provided)
P75	CS5#	PFCSE.CS5E = 1, PFCSS1.CS5S[1:0] = 01		(not provided)
P76	CS6#	PFCSE.CS6E = 1, PFCSS1.CS6S[1:0] = 01		(not provided)
P77	CS7#	PFCSE.CS7E = 1, PFCSS1.CS7S[1:0] = 01		(not provided)
P90	D16	PFBCR0.DH32E = 1	(not supported)	(not provided)
	A16	PFAOE1.A16E = 1, PFBCR0.ADRHMS = 1, PFBCR0.ADRHMS2 = 0		(not provided)
P91	D17	PFBCR0.DH32E = 1	(not supported)	(not provided)
	A17	PFAOE1.A17E = 1, PFBCR0.ADRHMS = 1, PFBCR0.ADRHMS2 = 0		(not provided)
P92	D18	PFBCR0.DH32E = 1	(not supported)	(not provided)
	A18	PFAOE1.A18E = 1, PFBCR0.ADRHMS = 1, PFBCR0.ADRHMS2 = 0		(not provided)
P93	D19	PFBCR0.DH32E = 1	(not supported)	(not provided)
	A19	PFAOE1.A19E = 1, PFBCR0.ADRHMS = 1, PFBCR0.ADRHMS2 = 0		(not provided)
P94	D20	PFBCR0.DH32E = 1	(not provided)	(not provided)
	A20	PFAOE1.A20E = 1, PFBCR0.ADRHMS = 1, PFBCR0.ADRHMS2 = 0	(not provided)	(not provided)
P95	D21	PFBCR0.DH32E = 1	(not provided)	(not provided)
	A21	PFAOE1.A21E = 1, PFBCR0.ADRHMS = 1, PFBCR0.ADRHMS2 = 0	(not provided)	(not provided)
P96	D22	PFBCR0.DH32E = 1	(not provided)	(not provided)
	A22	PFAOE1.A22E = 1, PFBCR0.ADRHMS = 1, PFBCR0.ADRHMS2 = 0	(not provided)	(not provided)
P97	D23	PFBCR0.DH32E = 1	(not provided)	(not provided)
	A23	PFAOE1.A23E = 1, PFBCR0.ADRHMS = 1, PFBCR0.ADRHMS2 = 0	(not provided)	(not provided)
PA0	A0	PFBCR0.ADRLE = 1, CSnMOD.WRMOD = 0		
	BC0#	PFBCR0.ADRLE = 1, CSnMOD.WRMOD = 1		
	DQM2	PFBCR0.ADRLE = 1, SDCCR.EXENB = 1, SDCCR.BSIZE[1:0] = 01	(not supported)	(not supported)
PA1	A1	PFBCR0.ADRLE = 1		
	DQM3	PFBCR0.ADRLE = 1, SDCCR.EXENB = 1, SDCCR.BSIZE[1:0] = 01	(not supported)	(not supported)
PA2	A2	PFBCR0.ADRLE = 1		
PA3	A3	PFBCR0.ADRLE = 1		
PA4	A4	PFBCR0.ADRLE = 1		
PA5	A5	PFBCR0.ADRLE = 1		
PA6	A6	PFBCR0.ADRLE = 1		
PA7	A7	PFBCR0.ADRLE = 1		
PB0	A8	PFAOE0.A8E = 1		
PB1	A9	PFAOE0.A9E = 1		
PB2	A10	PFAOE0.A10E = 1		
PB3	A11	PFAOE0.A11E = 1		
PB4	A12	PFAOE0.A12E = 1		

Table 23.36 How to Set the External Bus Interface (3/4)

Port	Output Signal	Settings of MPC Registers		
		224-Pin, 176-Pin	145-Pin, 144-Pin	100-Pin
PB5	A13	PFAOE0.A13E = 1		
PB6	A14	PFAOE0.A14E = 1		
PB7	A15	PFAOE0.A15E = 1		
PC0	A16	PFAOE1.A16E = 1, PFBCR0.ADRHMS = 0		
PC1	A17	PFAOE1.A17E = 1, PFBCR0.ADRHMS = 0		
PC2	A18	PFAOE1.A18E = 1, PFBCR0.ADRHMS = 0, PFBCR0.ADRHMS2 = 0		
PC3	A19	PFAOE1.A19E = 1, PFBCR0.ADRHMS = 0, PFBCR0.ADRHMS2 = 0		
PC4	A20	PFAOE1.A20E = 1, PFBCR0.ADRHMS = 0, PFBCR0.ADRHMS2 = 0		
	CS3#	PFCSE.CS3E = 1, PFCSS0.CS3S[1:0] = 10/11		
PC5	A21	PFAOE1.A21E = 1, PFBCR0.ADRHMS = 0		
	CS2#	PFCSE.CS2E = 1, PFCSS0.CS2S[1:0] = 10/11		
	WAIT#	PFBCR1.WAITS[1:0] = 10, PFBCR3.WAITS2 = 0		
	D3[A3/D3]	PFBCR2.D3S[1:0] = 10		
PC6	A22	PFAOE1.A22E = 1, PFBCR0.ADRHMS = 0		
	CS1#	PFCSE.CS1E = 1, PFCSS0.CS1S[1:0] = 10/11		
	D2[A2/D2]	PFBCR2.D2S[1:0] = 10		
PC7	A23	PFAOE1.A23E = 1, PFBCR0.ADRHMS = 0		
	CS0#	PFCSE.CS0E = 1, PFCSS0.CS0S = 1		
PD0	D0[A0/D0]	PFBCR2.D0S[1:0] = 00		
PD1	D1[A1/D1]	PFBCR2.D1S[1:0] = 00		
PD2	D2[A2/D2]	PFBCR2.D2S[1:0] = 00		
PD3	D3[A3/D3]	PFBCR2.D3S[1:0] = 00		
PD4	D4[A4/D4]	PFBCR3.DLHS = 0		
PD5	D5[A5/D5]	PFBCR3.DLHS = 0		
PD6	D6[A6/D6]	PFBCR3.DLHS = 0		
PD7	D7[A7/D7]	PFBCR3.DLHS = 0		
PE0	D8[A8/D8]	PFBCR0.DHE = 1		
	D0[A0/D0]	PFBCR2.D0S[1:0] = 01		
PE1	D9[A9/D9]	PFBCR0.DHE = 1		
	D1[A1/D1]	PFBCR2.D1S[1:0] = 01		
PE2	D10[A10/D10]	PFBCR0.DHE = 1		
	D2[A2/D2]	PFBCR2.D2S[1:0] = 01		
PE3	D11[A11/D11]	PFBCR0.DHE = 1		
	D3[A3/D3]	PFBCR2.D3S[1:0] = 01		
PE4	D12[A12/D12]	PFBCR0.DHE = 1		
	D4[A4/D4]	PFBCR3.DLHS = 1		
PE5	D13[A13/D13]	PFBCR0.DHE = 1		
	D5[A5/D5]	PFBCR3.DLHS = 1		
PE6	D14[A14/D14]	PFBCR0.DHE = 1		
	D6[A6/D6]	PFBCR3.DLHS = 1		
PE7	D15[A15/D15]	PFBCR0.DHE = 1		
	D7[A7/D7]	PFBCR3.DLHS = 1		
PF5	WAIT#	PFBCR1.WAITS[1:0] = 00, PFBCR3.WAITS2 = 1		(not provided)
PG0	D24	PFBCR0.DH32E = 1	(not provided)	(not provided)

Table 23.36 How to Set the External Bus Interface (4/4)

Port	Output Signal	Settings of MPC Registers		
		224-Pin, 176-Pin	145-Pin, 144-Pin	100-Pin
PG1	D25	PFBCR0.DH32E = 1	(not provided)	(not provided)
PG2	D26	PFBCR0.DH32E = 1	(not provided)	(not provided)
PG3	D27	PFBCR0.DH32E = 1	(not provided)	(not provided)
PG4	D28	PFBCR0.DH32E = 1	(not provided)	(not provided)
PG5	D29	PFBCR0.DH32E = 1	(not provided)	(not provided)
PG6	D30	PFBCR0.DH32E = 1	(not provided)	(not provided)
PG7	D31	PFBCR0.DH32E = 1	(not provided)	(not provided)

23.4 Usage Notes

23.4.1 Procedure for Specifying Input/Output Pin Function

Use the following procedure to specify the input/output pin functions.

- (1) Clear the pin mode register (PMR) for the target pin to 0 to select the general I/O port.
- (2) Specify the assignments of input/output signals for peripheral modules to the desired pins.
- (3) Enable writing to the Pmn pin function control register (PmnPFS) through the write-protect register (PWPR) setting. (m = 0 to 9, A to H, J to N, Q; n = 0 to 7)
- (4) Specify the input/output function for the pin through the PSEL[5:0] bit settings in the PmnPFS register.
- (5) Clear the PFSWE bit in the PWPR register to 0 to disable writing to the PmnPFS register.
- (6) Set the PMR to 1 as necessary to switch to the selected input/output function for the pin.

23.4.2 Notes on MPC Register Setting

- (1) Settings of the Pmn pin function control register (PmnPFS) should be made only while the PMR register for the target pin is set to 0. If the PmnPFS register is set while the PMR register is 1, unexpected edges may be input through the input pin or unexpected pulses are output through the output pin.
- (2) Only the allowed values (functions) should be specified in the Pmn pin function control registers. If a value that is not allowed for the register is specified, correct operation is not guaranteed.
- (3) Do not assign a single function to multiple pins through the MPC settings.
- (4) Ports 0, 4, 9, D, and E also function as analog Input/output pins for the A/D converter and D/A converter. When using these ports as analog Input/output pins, set them to general input pins by clearing the corresponding bits in the port mode register (PMR) and port direction register (PDR) to 0 and set the PmnPFS.ASEL bit to 1, to avoid degradation of accuracy.
- (5) The initial value of the time capture event input pin enable bit (TCEN) of the time capture control register y (RTCCRy, y = 0 to 2) is undefined after a reset. Therefore, set this bit to 0 to avoid unnecessary input.
- (6) Points to note regarding the port mode register (PMR), port direction register (PDR), and the PmnPFS register settings for pins that have multiplexed pin functions are listed in Table 23.37. The pin states are readable if the value of the ASEL bit is 0. Ensure that the PMR.Bj bit is 0 when changes to the PSEL[5:0] bits are made.

Table 23.37 Register Settings

Item	PMR.Bn	PDR.Bn	PmnPFS			Point to Note
			ASEL	ISEL	PSEL[5:0]	
After a reset	0	0	0	0	000000b	Pins function as general input port pins after release from the reset state.
General input ports	0	0	0	0/1	x	Set the PmnPFS.ISEL bit to 1 if these are multiplexed with interrupt inputs.
General output ports	0	1	0	0	x	
Peripheral functions	1	x	0	0/1	Peripheral functions (refer to Table 23.2 to Table 23.33)	Set the PmnPFS.ISEL bit to 1 if these are multiplexed with interrupt inputs.
Interrupt inputs	0	0	0	1	x	
NMI	x	x	x	x ^{*1}	x	Register settings are not required.
Analog inputs and outputs	0	0	1	x ^{*1}	x	Set these as general input port pins so that the output buffers are turned off.
Time-capture event-input pins	0	0	x	0/1	x	Set these as general input port pins so that the output buffers are turned off.
External bus	0	x ^{*2}	0	0	x	Set the PMR.Bn bit to 0 and do not select the peripheral function.
JTAG interface	0	x	x	0	x	Set the PMR.Bn bit and the PmnPFS.ISEL bit to 0 and switch the input buffers off.
FINE interface	0	x	x	0	x	Set the PMR.Bn bit and the PmnPFS.ISEL bit to 0 and switch the input buffers off.
EXTAL/XTAL	0	0	x	x ^{*1}	x	Set these as general input port pins so that the output buffers are turned off.

x: Setting not required.

0/1: Setting the PmnPFS.ISEL bit to 0 makes the pin incapable of functioning as an IRQ pin.

Setting the PmnPFS.ISEL bit to 1 makes the pin capable of functioning as an IRQ pin (if the IRQ is selected from the multiplexed functions).

Note 1. Even if the PmnPFS.ISEL bit is set to 1, the pin will not function as an IRQn input pin.

Note 2. To use the WAIT# input pin, set the corresponding bit in the PORTm.PDR register to 0.

- Note:
- The pin state is readable when the PmnPFS.ASEL bit is 0.
 - If the value of the PmnPFS.PSEL[5:0] bits is to be changed, do so while the PMR.Bn bit is 0.
 - If an RIIC function is assigned to a port pin, clear the PMR.Bn (to 0); pulling up is automatically turned off for outputs from peripheral modules other than the RIIC.
 - If an input pin for time-capture events is not in use, clear the time capture event input pin enable bit (TCEN) in time capture control register y (RTCCRY) (y = 0 to 2) to 0 (disabled). The value of the RTCCRY.TCEN bit after a reset is undefined.
 - Do not make settings to assign multiple external bus signals to a single pin.

23.4.3 Notes on the Use of Analog Functions

To use an analog function, set the corresponding bits in both the port mode register (PMR) and port direction register (PDR) to 0 so that the pin acts as a general input port. After that, set the pin function select bit in the Pmn pin function control register (PmnPFS.ASEL) to 1.